

# Modelling Financial Markets

## Chapter 12: Options and the Volatility Surface



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# What do option prices reveal about the future distribution?



Fischer Black. Photo: Dalmatine at English Wikipedia (2009 upload); public domain; Wikimedia Commons

*"If options are correctly priced in the market, it should not be possible to make sure profits by creating portfolios of long and short positions in options and their underlying stocks."*

— *Black & Scholes (1973)*

- Black–Scholes: replication gives a single implied volatility per asset
  - daily delta hedging, three-month call: error standard deviation 9.5% of the premium
- The market shows a smile: one-month implied volatility from the S&P 500 distribution
  - 30.6% at $K/S = 0.8$, against 13.9% at the money
- Bitcoin, 38-day expiry: risk-neutral probability of a fall of more than 30%
  - 1.24%, against 0.14% under the log-normal benchmark
- S&P 500, 1990–2026: mean VIX 19.4, mean realised volatility 15.3
  - variance risk premium: 4.09 vol points, positive on 86% of days

Central chart: implied vs realised volatility

# The Idea of the Chapter

- An option pays only in some states: its price is a bet on the **distribution**
  - not only on the direction
  - so it reveals what the market expects about volatility and crashes
- Part I, pricing by replication
  - parity, static spanning, the binomial tree
  - the Black–Scholes equation, the Greeks and the discrete-hedging error
- Then implied volatility, the smile and a full Bitcoin surface with its risk-neutral density
- Part II, implied volatility as information
  - variance swaps and the VIX (the Cboe index of 30-day S&P 500 implied volatility)
  - the variance risk premium and 0DTE options (zero days to expiry)
- Case studies: option-implied bounds on the equity premium and how the VIX reacts to volatility shocks
- AI (artificial intelligence) for scientific discovery
- **What you should be able to do**
  - derive put–call parity, the binomial price and the Black–Scholes equation
  - compute the Greeks, the discrete-hedging error and an arbitrage-free SVI surface with its density and bands
  - interpret option-implied forecasts (VIX, variance risk premium) with the right standard errors
  - Further Topics and Appendix: reading for familiarity

# Chicago, 1973: Listed Options Begin



Trading floor of the Chicago Board of Trade in 2006; its members founded the Cboe in 1973

Photo: Richie Diesterheft (2006); CC BY 2.0; Wikimedia Commons

- The Chicago Board Options Exchange (Cboe) opened in 1973 MacKenzie & Millo (2003)
  - the first exchange for standardised, listed options
  - Before: options were negotiated one by one, over the counter
- The same year: the Black–Scholes and Merton pricing papers Black & Scholes (1973), Merton (1973)
- Traders soon used the formula on the floor: theory changed the market it described MacKenzie & Millo (2003)

# The Formula and Its Authors



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons



Myron Scholes (2008)

Photo: Nobellaureatesphotographer (2008); CC BY-SA 3.0; Wikimedia Commons



Robert C. Merton (2006)

Photo: Digarnick (2006); public domain; Wikimedia Commons

- Key idea: a portfolio of the stock and cash can **replicate** the option Black & Scholes (1973)
  - so its price follows from no-arbitrage
- Scholes and Merton received the 1997 Nobel Memorial Prize in Economic Sciences
  - Black had died in 1995

## Put–Call Parity: What Does No Arbitrage Alone Imply?

- Two portfolios with the same pay-off at $T$ (European options, same $K$ and $T$, no dividends):
  - A: one call + cash $Ke^{-r\tau}$ (worth $K$ at $T$) $\Rightarrow \max(S_T, K)$
  - B: one put + one share $\Rightarrow \max(K, S_T)$
- Same pay-off in every state, so the same price today, otherwise there is an arbitrage:
  - $C_t - P_t = S_t - Ke^{-r\tau}$ (with dividend yield $q$: $S_t e^{-q\tau}$ instead of $S_t$)
- No model for $S$ is needed: parity uses only no-arbitrage Merton (1973)
- Consequence: a call and a put with the same $K$ and $T$ have the **same** implied volatility

## Static Replication: Any Pay-Off from a Bond, a Forward and Options

- Carr–Madan spanning: any twice-differentiable pay-off $f(S_T)$ and any point $F$:
  - Taylor with integral remainder; proof in the Appendix
  - $f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)\,(K - S_T)^+ dK + \int_F^\infty f''(K)\,(S_T - K)^+ dK$
  - Reading: a bond, a forward and $f''(K)\,dK$ puts below $F$ and calls above $F$
  - Priced today ($F$ the forward): $V_0 = e^{-r\tau} f(F) + \int_0^F f''(K)\,P(K)\,dK + \int_F^\infty f''(K)\,C(K)\,dK$
  - $P(K)$, $C(K)$: today's put and call prices at strike $K$, whatever the model for $S$ Bakshi, Kapadia & Madan (2003)
- Why this price: each piece has a market price today
  - the bond $e^{-r\tau}$ per unit, the forward 0, the options $P(K)$ and $C(K)$
  - No model for $S$ and no pricing measure are needed for $V_0$; the measure $Q$ of Section 3 is needed only to read option prices as expectations, as in the density bullet of the next slide

## Static Replication: Three Results Used Later

- "Risk-neutral": a price is the discounted mean pay-off under the pricing measure $Q$
  - not under real-world probabilities
  - $Q$ is constructed in Section 3
- **Variance**: $f(S) = -2\ln(S/F)$ has $f''(K) = 2/K^2$
  - the strip with weights $1/K^2$ is the variance swap and the VIX (Section 8)
  - Low strikes get the largest weights, so an expensive put wing makes variance expensive
- **Density**: a butterfly $[C(K-h) - 2C(K) + C(K+h)]/h^2 \to \partial^2 C/\partial K^2 = e^{-r\tau} q(K)$ Breeden & Litzenberger (1978)
  - $h$: the strike step; the risk-neutral density is a price (Section 7)
- Straddles, spreads and protective puts are finite strips
  - $f''$ concentrated at one or two strikes
  - moments and SVIX (simple VIX, the variance index of Martin (2017)): ▸ Appendix

# The Binomial Tree: How Do We Price an Option by Replication?

- One period of length $\Delta t$: $S \to uS$ or $dS$, with $d < R = e^{r\Delta t} < u$
  - $u$, $d$: up and down factors
  - $R$: gross riskless return
- Replicate the call with $\Delta$ shares and cash $B$: $\Delta uS + RB = C_u$ and $\Delta dS + RB = C_d$
  - $C_u$, $C_d$: the call values after an up and a down move
  - Solution: $\Delta = \dfrac{C_u - C_d}{(u-d)S}$, and the price is $C = \Delta S + B = \dfrac{q\,C_u + (1-q)\,C_d}{R}$
  - $q = \dfrac{R-d}{u-d} \in (0,1)$ (here a probability, not the dividend yield) is the unique probability under which $S_t/R^t$ is a martingale
- The real probability of an up move never enters
  - risk preferences are already in the price $S$
- **Fundamental theorem**: no arbitrage $\Leftrightarrow$ an equivalent martingale measure $Q$ exists Harrison & Pliska (1981), Delbaen & Schachermayer (1994)
  - $Q$ is unique exactly when the market is complete; with jumps or stochastic volatility there are many $Q$, and option prices carry risk premia (Section 8)

# The CRR Tree: $N$ Steps and Convergence to Black–Scholes

- Why $N$ steps: one period has only two final states
  - with $N$ steps the $N+1$ final states approximate the distribution of $S_T$
- The CRR (Cox–Ross–Rubinstein) tree Cox, Ross & Rubinstein (1979): $\Delta t = T/N$, $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$
  - This choice gives the one-step log return the variance $\sigma^2\Delta t$ (up to terms of order $\Delta t^2$), so the tree converges to the geometric Brownian motion of the Black–Scholes model
  - $q = (e^{r\Delta t} - d)/(u - d)$ is the same at every node
- European call: $C_0 = e^{-rT} \sum_{j=0}^{N} \binom{N}{j} q^j (1-q)^{N-j} \left(S_0 u^j d^{N-j} - K\right)^+$
  - $j$: the number of up moves; the sum is the discounted risk-neutral mean of the pay-off
- American options: value backwards from $T$
  - taking max(continuation value, exercise value) at every node

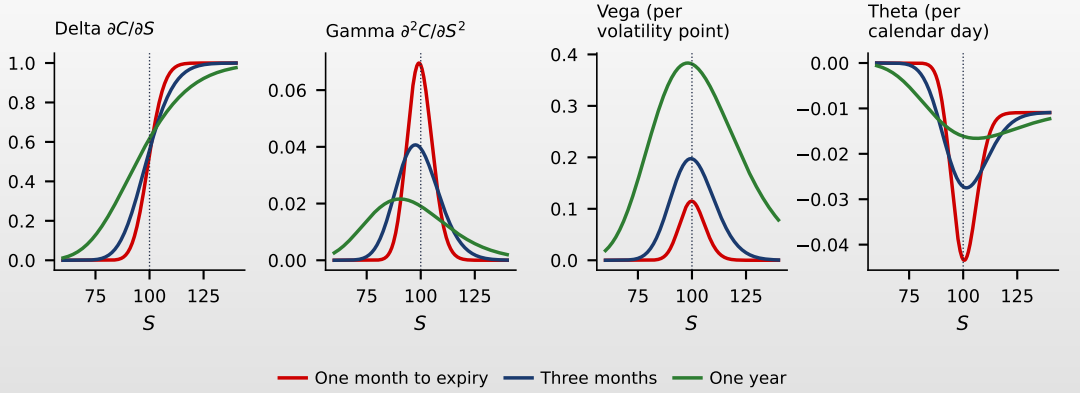
# The Black–Scholes Equation: What Price Does Continuous Replication Give?

- Geometric Brownian motion $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$
  - constant $\sigma$ and $r$
  - continuous trading, no costs, short selling allowed
- Replicate the option value $V(S,t)$ with a **self-financing** portfolio $\Pi$
  - $\Delta_t$ shares and cash $B_t$ at $r$: $d\Pi = \Delta\,dS + rB\,dt$
- Itô's lemma: $dV = \left(V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right)dt + \sigma S V_S\,dW_t$
  - $V_t$, $V_S$, $V_{SS}$: the partial derivatives
- Choose $\Delta = V_S$, start from $\Pi = V$ ($B = V - SV_S$)
  - $d(V-\Pi) = \left(V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS} - rB\right)dt$
  - both $dW_t$ and $\mu$ drop out
- A riskless position that costs nothing must earn nothing
  - hence the **Black–Scholes equation** Black & Scholes (1973), Merton (1973): $V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$
  - with $V(S,T) =$ pay-off
- That $\mu$ is irrelevant is a result, not an assumption
  - Feynman–Kac: $V = e^{-r\tau}E^Q[\text{pay-off}]$ with $S$ drifting at $r$ under $Q$

## The Black–Scholes Formula

- European call and put, no dividends (with yield $q$: replace $S$ by $Se^{-q\tau}$):
  - $C = S\,\Phi(d_1) - Ke^{-r\tau}\,\Phi(d_2), \qquad P = Ke^{-r\tau}\,\Phi(-d_2) - S\,\Phi(-d_1)$
  - $d_1 = \dfrac{\ln(S/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}$, with $\Phi$ the standard Normal distribution function
- Reading the two terms:
  - $\Phi(d_2)$ = risk-neutral probability that the call ends in the money
  - $\Phi(d_1)$ = number of shares in the replicating portfolio (the delta)
- Five inputs: $S$, $K$, $\tau$, $r$ observed
  - $\sigma$ is the only one not observed: it must be estimated or implied

# The Greeks: One Identity, All Sensitivities (1/2)




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# The Greeks: One Identity, All Sensitivities (2/2)

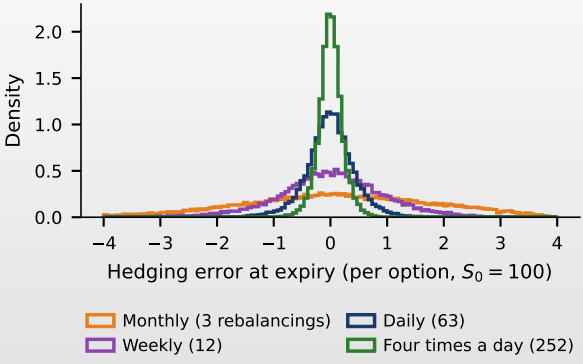
- Greeks: partial derivatives of the price
  - $\Delta = \partial C/\partial S$, $\Gamma = \partial^2 C/\partial S^2$
  - vega $\mathcal{V} = \partial C/\partial \sigma$, $\Theta = \partial C/\partial t$, rho $= \partial C/\partial r$
- Delta $\Delta = \Phi(d_1)$ (put: $\Phi(d_1) - 1$)
  - gamma $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$, $\varphi$ the standard Normal density
  - key identity $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- Vega $= S\varphi(d_1)\sqrt{\tau} = \Gamma S^2\sigma\tau$: for one expiry gamma and vega are proportional
  - Seminar 12, A1, A7
  - theta $\Theta = rC - rS\Delta - \frac{1}{2}\sigma^2 S^2\Gamma$
  - rho $= K\tau e^{-r\tau}\Phi(d_2)$
- Call with $K = 100$
  - short maturities concentrate gamma and theta at the strike
  - long maturities carry the most vega
  - vega per vol point, theta per calendar day

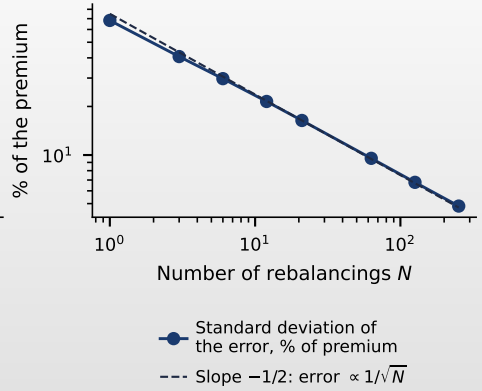
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## Discrete Delta Hedging: How Much Risk Is Left with $N$ Rebalancings?

- Black–Scholes replication needs continuous rebalancing
  - in practice the hedge is adjusted $N$ times, at $t_i = iT/N$
- At $t_0 = 0$: sell the call at its Black–Scholes price $C_0$
  - buy $\Delta_0 = \Phi(d_1)$ shares
  - keep the cash $B_0 = C_0 - \Delta_0 S_0$
- At $t_i$, $i = 1, \dots, N-1$: the portfolio is self-financing
  - $\Delta_i = \Phi\big(d_1(S_{t_i}, T - t_i)\big)$
  - $B_i = B_{i-1} e^{rT/N} - (\Delta_i - \Delta_{i-1}) S_{t_i}$
- **Hedging error** at expiry: $\varepsilon_N = \Delta_{N-1} S_T + B_{N-1} e^{rT/N} - (S_T - K)^+$
- Simulation: geometric Brownian motion with $\mu = 8\%$ and $\sigma = 20\%$
  - $S_0 = K = 100$, $r = 4\%$, $T = 0.25$
  - 20 000 paths; $N = 3, 12, 63, 252$
- Why $\mu \neq r$: replication never uses $\mu$
  - a real drift different from $r$ shows that the error stays centred at 0

# Discrete Delta Hedging: Algorithm and Error (1/2)




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# Discrete Delta Hedging: Algorithm and Error (2/2)

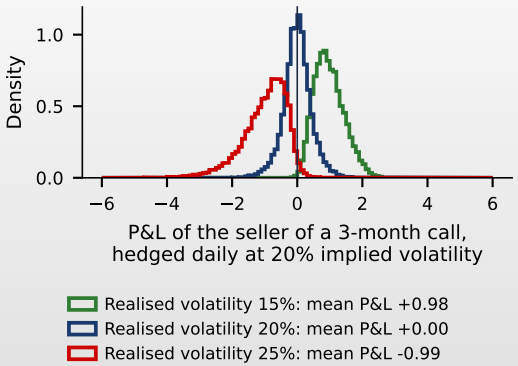
- Left: distribution of the error $\varepsilon_N$ for four rebalancing frequencies
  - right: standard deviation of the error, in % of the premium, against $N$ on log axes
- Black–Scholes premium 4.49 and $d_1 = 0.15$
- Mean error near 0
  - standard deviation (s.d.) in % of the premium: monthly 40.8%, weekly 21.5%, daily 9.5%, four times a day 4.8%
  - worst 1% of paths, in currency units per option: $-2.68$ (weekly), $-1.17$ (daily)
- Log-log slope $-0.48$
  - theory (Appendix): s.d. $\approx \sqrt{\pi/4}\,\mathcal{V}\sigma/\sqrt{N}$ ($\mathcal{V}$: vega)
  - theory 1.01, 0.44, 0.22 vs simulated 0.96, 0.43, 0.22 ($N = 12, 63, 252$)
- Transaction costs depend on the cost model
  - with a fixed cost per rebalancing, the total cost grows like $N$
  - with costs proportional to the traded value, in the standard diffusion setting, the expected cost grows roughly like $\sqrt{N}$

MFM_ch12_hedge_error

## Derivation: the Gamma-Weighted Volatility P&L

- Sell the option at $\sigma_{imp}$ and hold $\Delta = \partial_S V^{imp}$
  - $V^{imp}$: its Black–Scholes value at $\sigma_{imp}$
  - the price moves with $\sigma_{real}$: $dS = \mu S\,dt + \sigma_{real} S\,dW_t$
- Itô on $V^{imp}$: $dV = V_t\,dt + \Delta\,dS + \frac{1}{2}\Gamma\sigma_{real}^2 S^2 dt$
  - the equation at $\sigma_{imp}$ gives $V_t = rV - rS\Delta - \frac{1}{2}\sigma_{imp}^2 S^2\Gamma$
- Short option + $\Delta$ shares + cash at $r$, self-financing:
  - $d\,\text{P\&L} = r\,\text{P\&L}\,dt + \frac{1}{2}\,\Gamma_t S_t^2\left(\sigma_{imp}^2 - \sigma_{real}^2\right)dt$, so over the life
  - $\text{P\&L} = \displaystyle\int_0^T e^{r(T-t)}\,\frac{1}{2}\,\Gamma_t S_t^2\left(\sigma_{imp}^2 - \sigma_{real,t}^2\right)dt$: path-dependent, weighted by $\Gamma$, large near $K$ and near $T$
- A short variance swap pays $\sigma_{imp}^2 - \sigma_{real}^2$ with constant weights
  - a delta-hedged option is a variance bet with random weights
- First-order size: vega $\times$ (implied $-$ realised), checked on the next slide
  - Why: vega $= \Gamma S^2\sigma\tau$ (Section 4), so $\frac{1}{2}\Gamma S^2(\sigma_{imp}^2 - \sigma_{real}^2)\tau = \frac{1}{2}\Gamma S^2(\sigma_{imp} + \sigma_{real})(\sigma_{imp} - \sigma_{real})\tau \approx \Gamma S^2\sigma(\sigma_{imp} - \sigma_{real})\tau = \text{vega} \times (\sigma_{imp} - \sigma_{real})$

# Selling Volatility at 20%: Three Outcomes



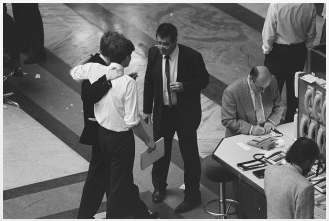

- Three-month call sold at 20% implied volatility, hedged daily
  - realised volatility 15%, 20% or 25%
- Realised 15%: mean P&L $+0.98$, positive on 99.8% of paths
- Realised 20%: mean $+0.00$; only discrete-hedging noise remains (s.d. 0.43)
- Realised 25%: mean $-0.99$, positive on only 2.2% of paths
- Each mean is close to vega $\times$ 5 vol points $= 0.197 \times 5 \approx \pm 0.99$
  - a clean bet on volatility

MFM_ch12_vol_mismatch ▶ Interactive

# Implied Volatility: Which $\sigma$ Explains the Market Price?

- **Implied volatility** $\sigma_{imp}$: the $\sigma$ that makes the Black–Scholes price equal the market price
  - $BS(S, K, \tau, r, \sigma_{imp}) = C^{market}$; $BS(\cdot)$: the Black–Scholes formula, $C^{market}$: the observed price
  - The solution is unique because vega $> 0$: the price increases in $\sigma$
  - There is no closed form: $\sigma$ enters both $\Phi(d_1)$ and $\Phi(d_2)$, so the equation is solved numerically
- Traders quote options in implied volatility, not in money
  - it is comparable across strikes, maturities and assets
- If Black–Scholes were right, all options on one asset would have the **same** implied volatility

# After October 1987: the Smile Appears



Amsterdam stock exchange after the crash, 21 October 1987

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- Before 1987 index options had a nearly flat implied volatility across strikes Rubinstein (1994)
  - Since the crash, low-strike puts are much more expensive: the **skew**
- October 1986 – August 1987: option prices showed crash fears Bates (1991)
  - Not strongly in the two months immediately before the crash
- **Smile**: implied volatility as a function of $K$
  - **term structure**: as a function of $T$
  - **surface**: both
- Crash risk on the Bucharest Stock Exchange: Pele & Mazurencu-Marinescu (2012)

## Why Smiles Exist and How Models Produce Them (1/2)

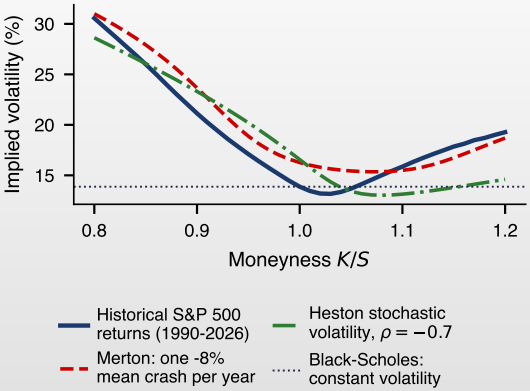
- **Fat tails**: large moves are more frequent than the Normal distribution allows (Chapter 1)
  - so deep OTM options are worth more than Black–Scholes says
- **Leverage and volatility feedback**: when prices fall, volatility rises Christie (1982)
  - negative skewness makes low-strike puts dearer
- **Crash fear**: the market prices a negative jump risk, visible in put prices Bates (1991)
- **Demand for insurance**: investors buy puts Gârleanu, Pedersen & Poteshman (2009)
  - intermediaries, who cannot fully hedge the risk, charge a premium for them

# Why Smiles Exist and How Models Produce Them (2/2)

| Model | Idea | Fits best |
|---|---|---|
| Jumps Merton (1976) | Poisson jumps added to the Brownian motion | steep short-dated smiles |
| Stochastic volatility Heston (1993), Hull & White (1987) | variance follows its own mean-reverting process, correlated with $S$ | skew and term structure of longer maturities |
| Local volatility Gatheral (2006) | $\sigma(S,t)$ read off the whole surface (the formula of Dupire (1994)) | matches every quoted price today, exotic options |
| Parametric smile, SVI (stochastic volatility inspired) Gatheral & Jacquier (2014) | five parameters per expiry, no dynamics | interpolation and quoting (Section 7) |

- Processes: Chapter 11
  - the implied-volatility surface in detail: Franke, Härdle & Hafner (2019), Čížek, Härdle & Weron (2011)

# Three Sources of a Smile (1/2)




MFM_ch12_smile_models ▶ Interactive

# Three Sources of a Smile (2/2)

- One-month implied volatility from three sources
  - the historical S&P 500 distribution, a jump model and a stochastic-volatility model
- History: 9 225 overlapping monthly returns
  - skewness $-1.27$, excess kurtosis 6.8
  - the excess kurtosis requires a finite fourth moment, which Chapter 1 questions
  - rescaled to a martingale ($r = 0$, no risk adjustment)
  - 30.6% at $K/S = 0.8$, 13.9% ATM, 19.3% at $K/S = 1.2$
- Merton jumps (one crash of about $-8\%$ a year): 31.0% at $K/S = 0.8$ vs 16.3% ATM
- Heston (price–variance correlation $\rho = -0.7$)
  - 28.6% at $K/S = 0.8$, 16.6% ATM, 14.6% at $K/S = 1.2$
- Risk-neutral laws are more left-skewed than realised ones Jackwerth (2000)
  - priced crash risk steepens market smiles further

MFM_ch12_smile_models

# The Bitcoin Surface: How to Describe the Smile Without Arbitrage?

- Deribit lists European options on Bitcoin, traded 24/7 Deribit DVOL
  - The options are **inverse**: premium and pay-off in bitcoin (BTC), not in US dollars (USD)
  - Each expiry has its own forward $F$
  - DVOL: Deribit's 30-day implied-volatility index
- Our snapshot: 22 September 2026, 07:51 UTC
  - 9 expiries that have at least two days left
  - 375 OTM options with a positive bid
- Why Bitcoin: the only asset in the course data with a full option chain
  - across strikes and expiries
- Crypto risk: Pele et al. (2023)
- Hedging Bitcoin options: Matic, Packham & Härdle (2023)

# The SVI Smile: Total Variance and Parameters

- Work with the **total implied variance** $w(k) = \sigma_{imp}^2(k)\,\tau$ at log-moneyness $k = \ln(K/F)$
- SVI parameterisation Gatheral & Jacquier (2014): $w(k) = a + b\left(\rho\,(k-m) + \sqrt{(k-m)^2 + s^2}\right)$
  - SVI: stochastic volatility inspired
  - $a$: level
  - $b$: slope of the wings
  - $\rho$: tilt (negative = put skew)
  - $m$: shift
  - $s$: curvature at the bottom

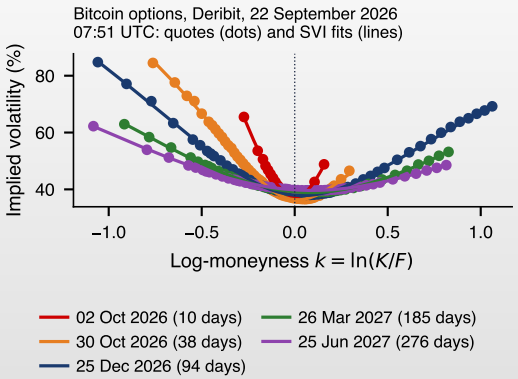
# The SVI Smile: Why This Shape and How We Fit It

- Why $w$ and $k$: the no-arbitrage conditions (next slide) are written directly in $w(k)$
  - different expiries become comparable
- Why SVI: as $|k| \to \infty$, $w$ becomes linear in $k$
  - the shape that Lee (2004) allows for the wings
  - five parameters are estimated from 24–58 quotes per expiry
- Wing slopes: $b(1-\rho)$ on the left and $b(1+\rho)$ on the right
- Fit, for each expiry: $\min \sum_j \left(w(k_j) - \sigma^2_{imp,j}\tau\right)^2$
  - least squares on $w$
  - with parameter bounds and six starting points
- Each fit is then checked against the no-arbitrage constraints

## No-Arbitrage Constraints on the Smile

- **Wings**: Lee (2004) bounds the asymptotic slope, $\limsup_{|k|\to\infty} w(k)/|k| \le 2$
  - for SVI $b(1+|\rho|) \le 2$ is necessary
  - $w \ge 0$ needs $a + bs\sqrt{1-\rho^2} \ge 0$
  - A steeper wing means fatter tails of $S_T$ under $Q$ (moment formula: Appendix)
- **Butterfly**: call prices convex in $K$, density $q \ge 0$ Gatheral & Jacquier (2014)
  - with $w' = \partial_k w$, $w'' = \partial_k^2 w$
  - $g(k) \ge 0$ is exactly a nonnegative density
  - $g(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{w'^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \ge 0$ for all $k$, plus $d_+(k) = -k/\sqrt{w} + \sqrt{w}/2 \to -\infty$ as $k \to \infty$ (Gatheral & Jacquier (2014), Lemma 2.2)
- **Calendar**: $\partial_\tau w(k,\tau) \ge 0$ for every $k$: total variance cannot fall with maturity
- Slice-by-slice least squares imposes none of them
  - SSVI, the surface version of SVI, has simple sufficient conditions (Gatheral & Jacquier (2014), Theorems 4.1–4.2)

# The Bitcoin Smile



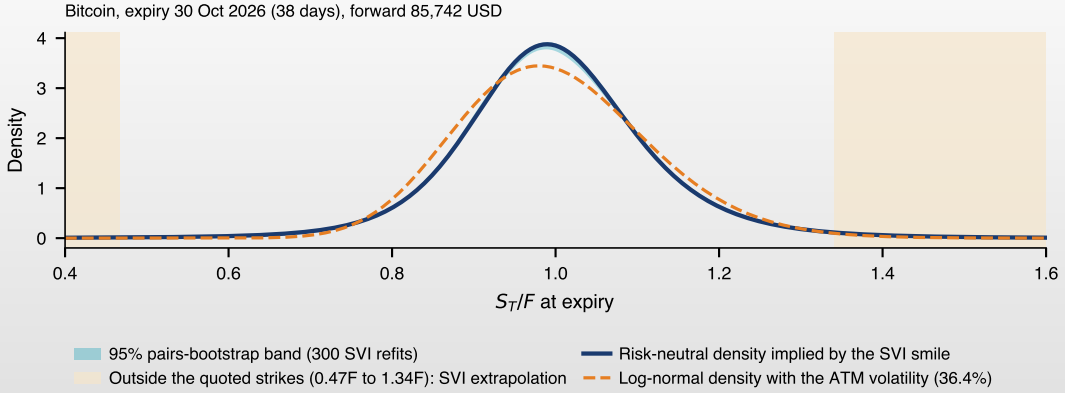

- Dots: mark IV of OTM options; lines: SVI fits, five expiries
- A true **smile**: both wings rise, unlike the one-sided equity skew
  - Bitcoin can jump up as well as down Zulfiqar & Gulzar (2021)
- Short expiries have the steepest wings
  - a jump over a few days weighs more relative to $\sigma\sqrt{\tau}$
- SVI root mean squared error: at most 0.70 vol points
- Puts dearer than calls at the same $|k|$: the skew is negative at all 9 expiries
- What the result supports: description

MFM_ch12_btc_smile ▶ Interactive

## The Risk-Neutral Density: From Prices to a Density

- From Breeden & Litzenberger (1978) (Section 2): $q(K) = e^{r\tau}\,\partial^2 C(K)/\partial K^2$
  - $q \geq 0$ is the butterfly condition above
- From noisy, discrete quotes this is an ill-posed inverse problem
  - differentiating twice amplifies the noise
  - Fit a smooth smile (SVI), then differentiate (other estimators: Appendix)
- Uncertainty: pairs bootstrap of the quotes, refit, differentiate
  - pointwise bands for $q$ and intervals for tail probabilities
- Beyond the last quoted strike the tail is an extrapolation
  - report where the quotes end
- $q$ mixes probabilities and risk aversion Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013)
  - pricing kernel $\propto q/p$, with $p$ the physical density
  - Its non-monotone shape is the pricing-kernel puzzle Jackwerth (2000)

# What the Market Prices for Bitcoin (1/2)




MFM_ch12_btc_rnd ▶ Interactive

# What the Market Prices for Bitcoin (2/2)

- Density from the SVI smile of the expiry closest to 30 days
  - 38 days, forward 85 742 USD, ATM 36.4%
- Band: 95% pairs bootstrap; benchmark: a log-normal density
  - shaded: outside the 56 quotes ($0.47F$ to $1.34F$)
- Probabilities: $F_Q(K) = 1 + \partial C/\partial K$, analytic in SVI (total mass 1, mean $F$)
  - bootstrap refits with butterfly arbitrage ($g(k) < 0$) are rejected: 5 of 305
- Fall of more than 30%: 1.24% $[1.16, 1.27]$ vs 0.14% log-normal
  - about 9 times more
  - 0.16 of the 1.24 points lie below the lowest quoted strike
- Fall of more than 20%: 4.35% vs 3.27%
  - rise above 20%: 5.3% in both
- Narrow band: it shows fitting noise, not bid–ask uncertainty
  - $\pm 2.0\%$ at the centre, $\pm 7\%$ at $0.7F$
  - mark volatilities are already smoothed
- Prices of insurance, not forecasts: they include the premium for crash risk
- What the result supports: description

MFM_ch12_btc_rnd

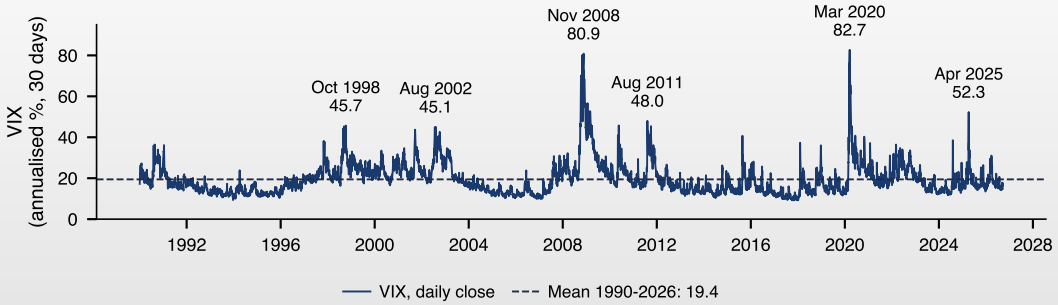
# Variance Swaps: What Does Future Variance Cost?

- A **variance swap** pays at $T$: notional $\times$ (realised variance $-$ strike $K_{var}$)
  - Realised variance $\frac{252}{n}\sum r_t^2$, $r_t$: the $n$ daily log returns to $T$
  - Fair strike: the risk-neutral expected variance Demeterfi, Derman, Kamal & Zou (1999)
  - $K_{var} = \frac{2e^{r\tau}}{\tau}\left(\int_0^F \frac{P(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\right)$ (spanning with $f = -2\ln(S/F)$, Section 2)
- Why it works: Itô on $\ln S$ (continuous paths, $r = q = 0$) Britten-Jones & Neuberger (2000)
  - $-2\ln(S_T/F) = \int_0^T \sigma_t^2\,dt - 2\int_0^T dS_t/S_t$
  - the last term is a $Q$-martingale
  - "Model-free" means free of a model for $\sigma_t$, not of continuity: with crash jumps the strip understates expected realised variance Carr & Wu (2009) (Appendix)
- Buyer of variance = insurance against turbulence Carr & Wu (2009)
  - seller = earns a premium, pays in crashes

# The VIX Index: the Cboe Formula

- The discrete version of the strip $K_{var}$ from the previous slide Cboe (2025):
  - $\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2$
  - $T$: time to expiry in years
  - $R$: risk-free rate
  - $K_i$: the strikes
  - $K_0$: the first strike at or below the forward index level $F$
  - $Q(K_i)$: bid–ask mid of the put for $K_i < K_0$, of the call for $K_i > K_0$, the average of both at $K_0$
  - The second term corrects for the strip being split at $K_0$ rather than exactly at $F$

# The VIX, 1990–2026



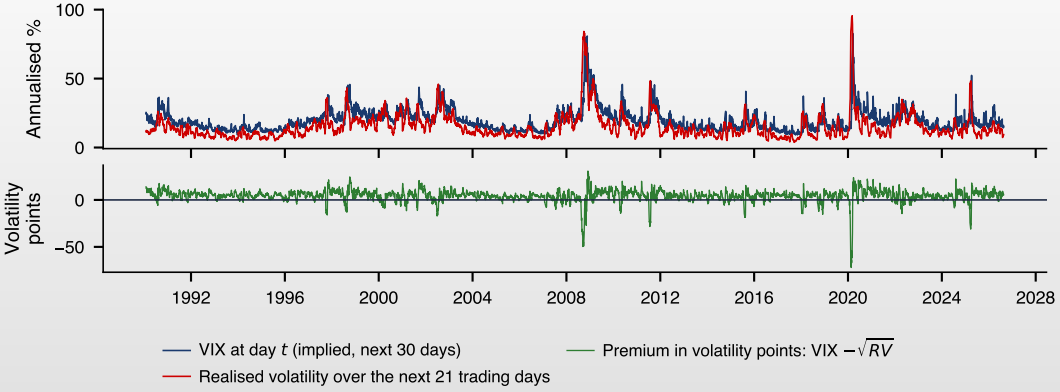

- Daily close with the six highest peaks at least 500 days apart
- Mean 19.4 (5.6% a month), range 9.1–82.7 (2020-03-16, COVID-19)
  - above 30 on 7.9% of days
  - sudden spikes, slow decay, as in GARCH (Chapter 5)

MFM_ch12_vix_history ▶ Interactive

# The Variance Risk Premium

- Variance risk premium (VRP) Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
  - three versions that must not be confused:
  - **Economic (ex ante)**: $E_t^Q[RV_{t,t+\mathbf{21}}] - E_t^P[RV_{t,t+\mathbf{21}}]$; $VIX_t^{\mathbf{2}}$ estimates the first term, the second needs a forecast
  - **Ex post** (used here): $VIX_t^{\mathbf{2}} - RV_{t,t+\mathbf{21}}$, the realised P&L of the seller of a variance swap, known only at $t+21$
  - **Ex-ante proxy**: $VIX_t^{\mathbf{2}} - RV_{t-\mathbf{21},t}$, past variance as the forecast of $E_t^P$; known at $t$, so it can serve as a predictor (Section 11)
- $RV_{t,t+\mathbf{21}} = \frac{\mathbf{252}}{\mathbf{21}} \sum_{i=\mathbf{1}}^{\mathbf{21}} r_{t+i}^{\mathbf{2}}$ (annualised, in $\%^{\mathbf{2}}$)
  - $r_{t+i}$: the daily log returns in %
  - roughly the same month as the 30 calendar days of the VIX
- Positive on average if investors dislike states with high volatility
  - these are also bad states for wealth
- Overlapping 21-day windows are autocorrelated: Newey–West with at least 21 lags Newey & West (1987)
  - 21 lags cover the overlap, not the persistence of volatility
  - standard error of the premium, in vol points: 0.27 with 21 lags, 0.31 with 42, 0.32 with 63, 0.33 with 126

# Implied vs Realised Volatility, S&P 500 (1/2)



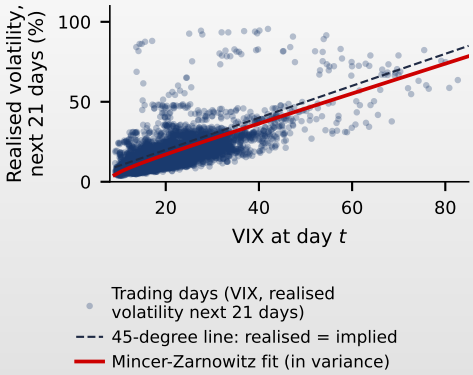

MFM_ch12_vrp ▶ Interactive

# Implied vs Realised Volatility, S&P 500 (2/2)

- Top: VIX and the volatility realised over the next 21 trading days
  - bottom: their difference VIX − RV, 1990–2026
  - RV $= \sqrt{RV_{t,t+\mathbf{21}}}$ in vol points
- 9 203 days: mean VIX 19.4, mean realised 15.3
  - Premium 4.09 vol points (Newey–West standard error, SE, 0.27); in variance, 113 $\%^{\mathbf{2}}$, $t = 5.5$
  - the $t$ in variance requires a finite fourth moment, which Chapter 1 questions
- Positive on 86% of days
  - the exceptions are crashes
  - Worst −72.0 vol points, window from 2020-02-20 (COVID-19)
- Same message as the delta-hedged option
  - sellers of volatility are paid on average and lose in crises
- What the result supports: description
  - an ex-post premium; the economic premium needs a forecast of $E_t^P$ (slide *The Variance Risk Premium*)

MFM_ch12_vrp

# Is the VIX a Good Forecast?



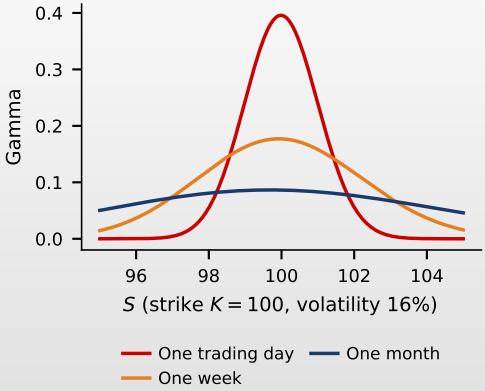

- Mincer–Zarnowitz Mincer & Zarnowitz (1969): $RV_{t,t+\mathbf{21}} = \alpha + \beta\, VIX_t^{\mathbf{2}} + \varepsilon_t$
  - Unbiased: $\alpha = 0$, $\beta = 1$; Newey–West errors, 21 lags
  - inference on coefficients needs $E\|x_t u_t\|^{\mathbf{2}} < \infty$ (the regression score) and weak dependence; in variance units, eighth moments of returns, which Chapter 1 questions
- $\hat{\alpha} = -51.9$ (34.4), $\hat{\beta} = 0.860$ (0.105), $R^{\mathbf{2}} = 0.37$
  - Separately neither differs from $(0, 1)$: $t = -1.51$ and $-1.34$
- But the estimates correlate $-0.90$: unbiasedness rejected
  - joint Wald $\chi^{\mathbf{2}}_{\mathbf{2}} = 41.7$, $p = 8.9 \cdot 10^{-\mathbf{10}}$
- Chart axes in volatility (%)
  - the fit estimated in variance drawn back in volatility
  - most points lie below the 45-degree line
- Past realised variance alone: $R^{\mathbf{2}} = 0.29$
  - With both: VIX keeps its weight (0.77), past variance adds little (0.08, SE 0.13)

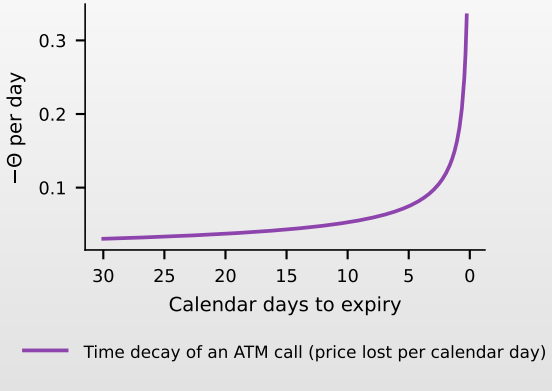
MFM_ch12_vix_rv_mz ▶ Interactive

## 0DTE Options: What Changes When Expiry Is Today?

- **0DTE** options are traded on the day they expire Cboe (2023)
  - S&P 500 options now expire every trading day
- More than 75% of retail trades in S&P 500 options involve 0DTE contracts Beckmeyer, Branger & Gayda (2023)
  - retail investors lose money on them
- 0DTE options were 48% of index-option trading in August 2023 Brogaard, Han & Won (2023)
- More 0DTE trading is followed by higher index volatility Brogaard, Han & Won (2023)
  - mostly through speculative retail trades
- Ultra-short pricing needs skewness and kurtosis corrections Bandi, Fusari & Renò (2023)
- From Section 4: as $\tau \to 0$, gamma and theta explode at the money

# Gamma and Time Decay Near Expiry (1/2)




MFM_ch12_0dte_gamma ▶ Interactive

# Gamma and Time Decay Near Expiry (2/2)

- Left: gamma of a call with $K = 100$ one day, one week and one month before expiry
  - right: daily time decay of an ATM call
- Inputs: $S = K = 100$, $\sigma = 16\%$, $r = 0$
  - gamma counts trading days ($\tau = 1/252$ and $21/252$)
  - time decay counts calendar days ($\tau = 1/365$ and $30/365$), as on the two axes
- ATM gamma one day before expiry 0.396 vs 0.086 one month before
  - 4.6 times larger ($= \sqrt{21}$)
  - a 1% move in the last day changes delta by about 0.4
- Time decay 0.167 in the last day vs 0.030 with 30 days left
  - about 5.5 times more
  - the seller collects fast decay and carries large gamma risk
  - The ATM value, about $0.4\, S\sigma\sqrt{\tau} = 0.33$, is lost entirely during that day, roughly twice the starting rate

MFM_ch12_0dte_gamma

# Case Study: the Research Question (1/2)

- **Questions** Back, Crotty & Kazempour (2022): option prices give lower bounds on the equity premium
  - Are these bounds valid conditionally?
  - Are they tight conditionally?
  - Do they forecast returns out of sample?
- Why: a tight bound is a real-time equity-premium forecast with no estimated parameter
  - regression of realised returns on the bound
  - Martin (2017) could not reject $\alpha = 0$, $\beta = 1$ on 1996–2012
  - a test with low power

# Case Study: the Research Question (2/2)

- **Martin bound** (eq. 3):
  - $R_{t,T}$ the gross market return, $R_{f,t,T}$ the gross riskless return
  - $\mathrm{var}^*_t$ the risk-neutral variance ($\mathrm{SVIX}^2_{t,T}$ not annualised here)
  - $E_t[R_{t,T}] - R_{f,t,T} \geq R_{f,t,T}\,\mathrm{SVIX}^2_{t,T}, \qquad \mathrm{SVIX}^2_{t,T} = \mathrm{var}^*_t(R_{t,T}/R_{f,t,T})$
  - Valid if $\mathrm{cov}_t(M_{t,T}R_{t,T}, R_{t,T}) \leq 0$ ($M_{t,T}$ the stochastic discount factor); tight if this covariance is zero, e.g. a log-utility investor who holds the market
- **Chabi-Yo–Loudis bound** Chabi-Yo & Loudis (2020): built from the first four risk-neutral moments
  - free of preference parameters in its restricted form
- **Link to this chapter**: SVIX is the static-spanning strip with $f = S^2$
  - Section 2 and ▸ Appendix
  - Realised slackness $s_{t,T} = R^e_{t,T} - b_{t,T}$ (excess return minus bound): valid means $E_t[s_{t,T}] \geq 0$, tight means $E_t[s_{t,T}] = 0$

# Case Study: Data, Moments and Forecasting

- **Data** (Section 3 and Appendix C of the paper)
  - filters: Appendix
  - Daily Martin and Chabi-Yo–Loudis bounds, January 1990 – December 2020, from S&P 500 index options, interpolated to 30, 90, 180 and 360 days
  - S&P 500 excess returns over 21, 63, 126 and 252 trading days, annualised, in %
  - compounded as $\prod(1 + r - r_f) - 1$, the convention of the authors' code; it differs from $\prod(1 + r) - \prod(1 + r_f)$
- **Moments** (eq. 10): $E[(R^e_{t,T} - b_{t,T})\, z_t] \geq 0$
  - for any nonnegative $z_t$ known at $t$
  - $z_t$: a constant and nine Welch–Goyal predictors made positive Welch & Goyal (2008) (Table 3)
- **Forecasting**: bias-corrected predictive regressions Amihud & Hurvich (2004) (Tables 5, 6)
  - Amihud–Hurvich: a persistent predictor biases the OLS slope (Stambaugh bias), so the innovation of the predictor, estimated from a bias-corrected AR(1), is added as a regressor
- Monthly out-of-sample forecasts against the historical mean
  - measured by $R^2_{OS}$, the out-of-sample $R^2$ against that mean
  - Table 8; protocol: Appendix

## Case Study: the Kodde–Palm Tests

- Kodde & Palm (1986) (eqs. 11–14): $\bar{\lambda}$ is the vector of the 10 sample means
  - $\Sigma$: their Hansen–Hodrick covariance Hansen & Hodrick (1980)
  - with lags = days in the horizon
  - The Hansen–Hodrick covariance keeps, with equal weights, exactly the autocovariances created by the overlap of the return windows
- $D_1 = \min_{\lambda \geq 0}(\lambda - \bar{\lambda})^\top \Sigma^{-1}(\lambda - \bar{\lambda})$ tests validity
  - $H_0$: $\lambda_0 \geq 0$
  - the distance to the region where all moments are $\geq 0$
  - zero if all are already positive
- $D_2 = \bar{\lambda}^\top \Sigma^{-1} \bar{\lambda} - D_1$ tests tightness ($H_0$: $\lambda_0 = 0$ vs $\lambda_0 \geq 0$): a large $D_2$ rejects tightness
  - Example with two moments and $\Sigma = I$: $\bar{\lambda} = (0.5, -0.2)$ gives $D_1 = 0.2^2 = 0.04$ and $D_2 = 0.29 - 0.04 = 0.25$; with $\bar{\lambda} = (0.5, 0.2)$, $D_1 = 0$ and $D_2 = 0.29$
- Under $H_0$ some inequality constraints bind and others do not
  - both statistics follow weighted mixtures of $\chi^2_0, \ldots, \chi^2_{10}$
  - bounds on the critical values tabulated by Kodde & Palm (1986)

# Case Study: Results, Validity and Tightness (Tables 1 and 3)

| Horizon | Mean $R^e$ (%) | Mean $b$ (%) | $D_1$ | $D_2$ | $\hat{\beta}$ (SE) | $R^2_{OS}$ (%) |
|---|---|---|---|---|---|---|
| 1 month | 8.76 | 3.81 | 0.00 | 29.93 | 1.36 (1.23) | −1.2 |
| 3 months | 8.73 | 3.86 | 0.00 | 39.51 | 1.39 (1.47) | −1.2 |
| 6 months | 8.67 | 3.83 | 0.00 | 49.28 | 1.78 (0.85) | −0.2 |
| 12 months | 8.86 | 3.68 | 0.00 | 71.21 | 1.04 (0.88) | −3.0 |

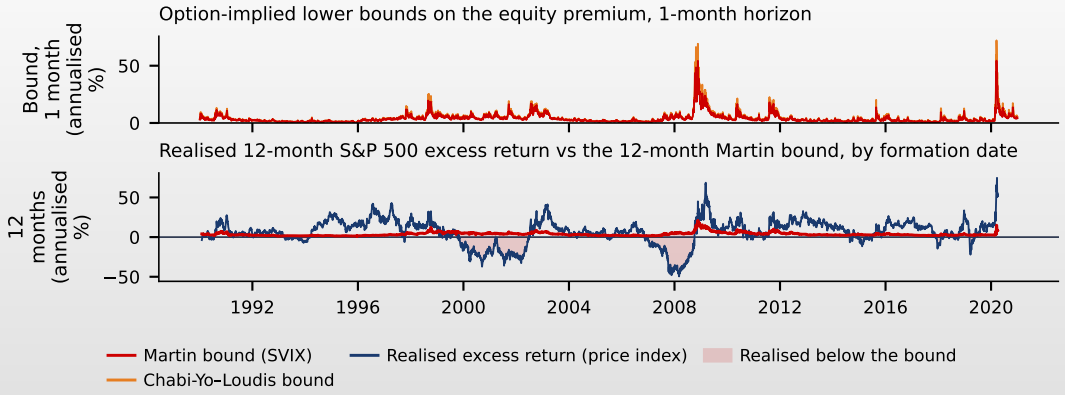
- Source: Back, Crotty & Kazempour (2022), *Journal of Financial Economics*, Tables 1, 3, 5, 6 and 8
- **Validity not rejected**: $D_1 = 0$ and $p = 100\%$ at every horizon
  - all 10 sample moments are positive
  - this does not prove validity in every state
- **Not tight**: $D_2$ exceeds 28.86 at every horizon
  - the upper 0.1% Kodde–Palm critical value for 10 restrictions
  - finite-sample $p$: 3.6, 0.4, 2.6, 8.3%
  - mean slackness 4.8 to 5.2 points a year: the bound is less than half of the realised premium
- What the result supports: description
  - mean slackness over 1990–2020, not forecasting power: out of sample the bound loses to the historical mean (Further Topics)


▸ Further Topics: forecasting results

# Case Study: Do the Two Tightness Tests Agree?

- Same $D_2$, two yardsticks
  - the asymptotic critical value rejects tightness at every horizon
  - the finite-sample $p$-values reject at 5% at 1, 3 and 6 months, but not at 12 months
- With a Holm correction for four horizons (Section 11), the finite-sample test rejects only at 3 months
  - $0.4\% < 1.25\%$, then $2.6\% > 1.67\%$
- The asymptotic test rejects at every horizon even after the correction
- Why they can disagree: $\Sigma$ is imprecise
  - 1990–2020 holds only about 31 non-overlapping 12-month windows
  - the moments use persistent predictors
  - so asymptotic critical values can be too small
- The finite-sample $p$ is the more cautious reading
  - report both, never only the one that rejects (Seminar 12, C3)
- The mean slackness of 4.8–5.2 points a year does not depend on the test
  - economically, the bound covers less than half of the realised premium at every horizon

# Case Study: the Bound and the Realised Premium (1/2)




MFM_ch12_bck_bounds ▶ Interactive

# Case Study: the Bound and the Realised Premium (2/2)

- Top: the authors' daily 1-month bounds, 1990–2020 (replication package)
  - bottom, course data: 12-month excess return of the S&P 500 price index
  - compared with the 12-month Martin bound
- 1-month bound: median 2.7%, peak 54.4% (16 March 2020)
  - 12-month bound: peak 21.1% (20 November 2008)
  - A risk-neutral variance, the bound jumps with the VIX in crises
- Chabi-Yo–Loudis $\geq$ Martin on 99.99% of days
  - negative skewness and fat tails raise it
- Realised 12-month excess return below the bound on 33% of dates (2000–2002, 2007–2008)
  - No contradiction: the bound restricts $E_t[R^e_{t,T}]$, not each realisation
- Correlation bound–realised 0.15 (overlapping, descriptive only)
  - return noise dwarfs the bound
  - hence imprecise slopes and negative $R^2_{OS}$

MFM_ch12_bck_bounds

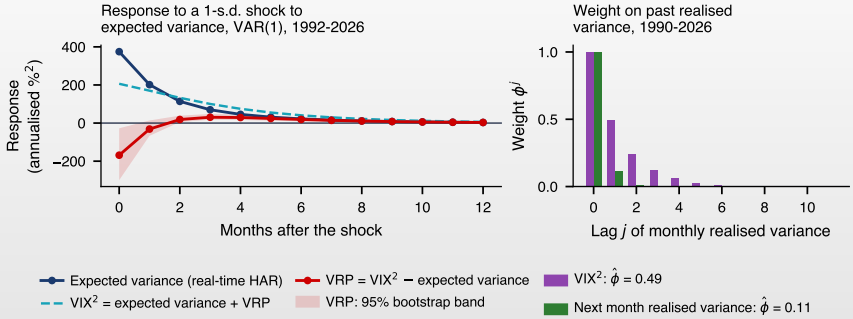
# Case Study: Replication and Limitations

- **Replicate** (Seminar 12, C3)
  - the authors' daily bound series (replication package)
  - the S&P 500 daily closes of the course data
  - The daily risk-free rate (Kenneth French Data Library) and the Welch–Goyal predictors (Amit Goyal's data page)
  - Rebuild Table 1 and $D_1$, $D_2$ of Table 3
- **Limitations stated by the authors**
  - The series starts in 1990 because the bounds need option prices: the out-of-sample tests have little power (paper, Section 6)
  - Validity holds even when the bound is replaced by its time-series mean, so it says nothing about predictive power (paper, Section 4.3)
  - More recent bounds and formulas for risk premia are not tested (paper, Introduction)

## Does the VIX Underreact to Volatility?

- Lochstoer & Muir (2022): investors update their expectations of market volatility too slowly
  - underreaction first, after a volatility shock, then delayed overreaction
- Why it matters: $\text{VIX}^2$ is the risk-neutral price of 30-day variance
  - if the VIX rises less than expected variance, the VRP (variance risk premium) falls exactly when risk rises
  - VRP = $\text{VIX}^2$ minus the objective expected variance $\hat{\sigma}_t^2$
- The market's expectation as a weighted average of the past (equation 4 of the paper): $\text{VIX}_t^2 = a + b \sum_{j=0}^{11} \phi^j \, rv_{t-j} + \varepsilon_t$
  - $rv_t$: realised variance of month $t$, the sum of squared daily S&P 500 log returns, annualised
  - $\phi = 0$: only the current month counts; a larger $\phi$ means expectations that move more slowly
- The paper's numbers (Journal of Finance, Section I): $\phi = 0.42$ (standard error 0.12) for $\text{VIX}^2$, January 1990 to April 2020
  - after a shock to expected variance, the VRP response is negative at first and changes sign after the third month (Figure 2)
- The consequence the authors propose: a weak risk–return tradeoff, because the premium rises only once volatility has started to fall

# How the VIX Responds to a Shock to Expected Variance




- **Left: the monthly response, in annualised %², to a one-standard-deviation shock to $\hat{\sigma}^2$ (VAR, 1992–2026)**
- **Right: the weights $\hat{\phi}^j$ on the variance of month $t-j$, for VIX² and for next month's realised variance**

MFM_ch12_vol_expectations ▶ Interactive

## The Paper and the Course Replication

- The paper: $\phi$ estimated from equation 4 for $\text{VIX}^2$ and, as the rational benchmark, for subsequent realised variance
  - $\hat{\sigma}_t^2$: the real-time forecast of next month's variance, a HAR (heterogeneous autoregressive) model on 5-minute returns
  - Table III: regressions on $\hat{\sigma}_t^2$ and on the weighted average of the last 6 values of $\hat{\sigma}^2$, with weights proportional to $0.5^j$
  - a monthly VAR(1) (vector autoregression) with $\hat{\sigma}^2$ ordered first (Cholesky) gives the responses to the shock
- The course replication: daily S&P 500 and VIX, January 1990 to Aug 2026 (440 months)
  - $\hat{\sigma}_t^2$: HAR on daily returns (last day, last 5 and last 21 days), re-estimated each month on the data available then
  - the first 24 months are the burn-in, so $\hat{\sigma}^2$ starts in January 1992
  - the VAR has three variables: $\hat{\sigma}^2$, the VRP and the excess return of the price index (risk-free rate: Kenneth French)
  - the paper adds the log price–dividend ratio, not in the course data
- Inference
  - standard errors of $\hat{\phi}$: moving block bootstrap with 18-month blocks, like the paper's 6 quarters
  - Table III: Newey–West standard errors with 12 lags; VAR bands: residual bootstrap
  - HAC coefficients need $E\|x_t u_t\|^2 < \infty$ and weak dependence; with $x_t$ and $u_t$ in variance units, eighth moments of returns, which Chapter 1 questions

## The VIX Depends More Persistently on Past Variance

- $\hat{\phi}$ for VIX$^2$: 0.54 (standard error 0.08) on the paper's sample, 0.49 (0.08) up to 2026
  - for next month's realised variance: 0.26 (0.30) and 0.11 (0.20), not significantly different from zero
- Table III, 334 months, 1992–2020: next month's variance depends on $\hat{\sigma}_t^2$ (0.82, s.e. 0.12); the past-average coefficient is not significant (0.04, s.e. 0.10)
  - in the paper (Panel A, column 6): 1.53 (0.54) and $-0.58$ (0.46)
- VIX$^2$ depends on both: 0.56 (0.09) and 0.41 (0.04), $R^2 = 0.79$
  - in the paper (Panel A, column 7): 0.30 (0.12) and 0.64 (0.07), $R^2 = 0.79$
- VAR: a shock of 375 %$^2$ to $\hat{\sigma}^2$ raises VIX$^2$ by only 206 %$^2$ (55% of the shock)
  - VRP response: $-169$ %$^2$ on impact (pointwise 95% band: $[-300, -27]$), positive from month 2, $+30$ in month 3
- What the result supports: a description consistent with slow expectations, on daily instead of 5-minute data
  - it does not identify the mechanism: a rationally time-varying price of variance risk could give the same signs
- The limits
  - the daily HAR forecasts less well: $R^2 = 0.32$ for next month's variance, against 0.47 in the paper
  - part of the VIX weight on the past may be information our HAR lacks
  - months of very high variance (2008, 2020) dominate the regressions in levels; the Cholesky ordering changes the initial sign (next slide)

## Inference in the Case Study: What the Bands Cover

- The difference $\hat{\phi}_{\mathrm{VIX}} - \hat{\phi}_{RV}$: bootstrap on the same 18-month blocks for both regressions
  - paper's sample: 0.29, 95% interval $[-0.05, 0.91]$, so the difference is not significant at 5%
  - up to 2026: 0.38, 95% interval $[0.08, 0.80]$, significant at 5%
- The VAR bands are conditional on the estimated $\hat{\sigma}^2$ series
  - the bootstrap resamples the VAR residuals without re-estimating the first-stage HAR
  - the bands are pointwise, horizon by horizon, not simultaneous over all horizons
  - the Newey–West errors of Table III also leave out the uncertainty of $\hat{\sigma}^2$
- VAR(1) diagnostics
  - stable: the largest eigenvalue has modulus 0.73
  - the return residuals are conditionally heteroskedastic (Engle test, 12 lags, $p < 0.001$)
  - resampling the residuals independently is therefore only an approximation
- The Cholesky ordering matters
  - with $\hat{\sigma}^2$ ordered last, the VRP response is 0 on impact, by construction, and $+33$ %$^2$ in month 1
  - the negative initial sign rests on $\hat{\sigma}^2$ not responding within the month to VRP shocks

## The Section's Question and the Tools to Test It

- The section's question: does the VRP (Section 8) still predict equity returns after 2007?
- **Hodrick 1B** Hodrick (1992): keep the regression of overlapping $h$-month returns on $\mathrm{VRP}_t$
  - only the covariance estimator of the coefficient changes
  - its score: the one-month residual under $H_0$ times the sum of the last $h$ values of the regressor
  - so the covariance needs no autocovariances of overlapping errors
- $R^2_{OS} > 0$: the predictive model beats the historical mean
  - lower mean squared forecast error than the historical mean
  - **Clark–West** Clark & West (2007) tests this for nested models
  - it adjusts for the noise that estimating a truly zero slope adds to the larger model
- **Holm correction** for $m$ tests (e.g. four horizons)
  - sort the $p$-values
  - compare the smallest with $\alpha/m$, the next with $\alpha/(m-1)$, and so on
  - stop at the first that fails
  - Example: $p = 0.01, 0.03, 0.04$, $\alpha = 0.05$, $m = 3$: $0.01 < 0.0167$ rejects, $0.03 > 0.025$ stops, so only one hypothesis is rejected, although all three $p < 0.05$

# An Open Question: Does the VRP Still Predict Returns?

- Bollerslev, Tauchen & Zhou (2009), Table 2: monthly data 1990–2007
  - $y_t^{(h)} = \frac{12}{h} \sum_{j=1}^{h} (r_{m,t+j} - r_{f,t+j})$ on $\mathrm{VRP}_t = \mathrm{VIX}_t^2/12 - \mathrm{RV}_t$
  - $\mathrm{RV}_t$: realised variance of month $t$; both terms in monthly, not annualised, $\%^2$ (a factor 12 below Section 8)
  - At $h = 3$: slope 0.47, Hodrick $t = 2.86$, adjusted $R^2 = 6.82\%$, the peak of a hump-shaped horizon profile
  - the Hodrick $t$ on a variance regressor requires a finite fourth moment, which Chapter 1 questions
- Goyal, Welch & Zafirov (2024): 29 equity-premium predictors proposed after 2008, re-examined with data to 2021
  - over one third lose in-sample significance
  - half of the rest fail out of sample
- Open: is VRP predictability a stable price of variance and jump risk, or a property of 1990–2007?
  - the 2008–2026 sample adds two crashes and the 0DTE era
- Formally: $H_0 : \beta_3 = 0$ vs $H_1 : \beta_3 > 0$ in $y_t^{(3)} = \alpha + \beta_3 \mathrm{VRP}_t + u_t$
  - Out of sample: $H_0$: the VRP forecast does not lower the mean squared prediction error of the historical mean; Clark–West statistic Clark & West (2007)

# The Discovery Loop and What the Researcher Must Check (1/2)

- **Literature**: ask an assistant for peer-reviewed tests of VRP predictability since 2009
  - DOI (digital object identifier), sample, table
  - Check: resolve every DOI on Crossref and compare the cited table with the paper, digit by digit
- **Hypotheses**: three candidate explanations
  - time-varying risk aversion
  - jump-risk compensation
  - or a small-sample artefact of persistent regressors and overlapping returns
  - Check: each must imply something testable for the horizon profile, for 2008–2026 and for non-US markets

# The Discovery Loop and What the Researcher Must Check (2/2)

- **Data and code**: build $\text{VIX}^2/12$ and monthly RV from the course data
  - from the daily VIX and S&P 500 closes
  - reproduce Table 2 ($h = 3$) before extending anything
  - Check timing (VIX at the end of month $t$, returns from $t + 1$) and match the published means (implied 33.23, realised 14.93)
- **Robustness and critique**: a dated pre-analysis plan
  - horizons, start date, benchmark, Holm correction
  - a hostile-referee prompt
  - Check overlap (Hodrick errors Hodrick (1992)), horizon and start-date mining ($R^2_{OS}$ against the start date), the weight of 2008 and 2020; full prompts: Appendix

# Mini-Case: Replication and Extension (1/2)

- A typical assistant proposal, tested on the course data
  - "the VRP predicts the quarterly equity premium; extend the 2009 result to today"
- Replication, February 1990 to September 2007, $h = 3$, daily data
  - $\hat{\beta}_3 = 0.42$, $t_{\text{Hodrick}} = 2.25$, adjusted $R^2 = 4.6\%$
  - published: 0.47, 2.86, 6.82%
  - Mean implied variance 33.1 is close to 33.23; mean realised 20.6 vs 14.93: daily close-to-close data
- Extension to May 2026 (436 months)
  - $\hat{\beta}_3 = 0.09$, $t_{\text{Hodrick}} = 1.33$, adjusted $R^2 = 1.0\%$
  - all horizons with four kinds of standard errors: Appendix

# Mini-Case: Replication and Extension (2/2)

- Out of sample from January 2008: 221 expanding-window forecasts
  - $R^2_{OS} = 1 - \sum_t (y_t - \hat{y}_t)^2 / \sum_t (y_t - \bar{y}_t)^2 = -22.2\%$ Campbell & Thompson (2008); $\bar{y}_t$: historical mean up to $t$
  - Clark–West $t = 0.64$ ($p = 0.26$): $H_0$ not rejected
  - the Clark–West test on squared errors requires a finite fourth moment, which Chapter 1 questions
- Reading: sign and size replicate in-sample
  - the longer sample does not support stable predictability
  - outliers, measurement or a vanished premium: that is the open part
- What the result supports: comparison of procedures
  - the VRP forecast against the historical mean

MFM_ch12_ai_discovery

# Your Turn: a Project on the VRP After 2008 (1/2)

- Questions
  - Does the VRP predict the equity premium after 2008?
  - Does the Bitcoin premium of Section 8 (DVOL) predict Bitcoin returns?
- Replicate first: Bollerslev, Tauchen & Zhou (2009), Table 2, $h = 3$
  - the published numbers of the open question
  - explain every gap

# Your Turn: a Project on the VRP After 2008 (2/2)

- Deliverables
  - A dated pre-analysis plan; the replication table; out-of-sample results for all horizons with a Holm correction
  - Robustness (subsamples, without 2008 and 2020; optionally 5-minute realised variance, if you have intraday data); a referee-style critique and your reply
- Log each prompt that shaped a decision in `AI_USE.md`
  - log each invented reference, wrong number or look-ahead caught in `AI_ERRORS.md`
  - with the test that caught it (templates)

## What Option Prices Measure, and Their Limits

- Parity, bounds and static spanning need no model
  - the tree and the Black–Scholes equation price by dynamic replication
- Discrete hedging leaves an error of order $1/\sqrt{N}$
  - a delta-hedged option is a bet on realised vs implied variance
- Implied variance exceeded realised variance on average, on the S&P 500 and on Bitcoin
- The surface and the risk-neutral density are estimates
  - check the no-arbitrage constraints, reject estimates that violate them and report bands
- The VIX is a strip estimator with jump and truncation biases
  - as a forecast it needs joint tests, logs and HAC loss comparisons
- Option-implied equity-premium bounds: validity not rejected for the moments and sample tested; not tight in 1990–2020
  - evidence rests on average slackness, not on forecasting power
- "Not tight" is robust on the asymptotic test
  - less certain on the finite-sample tests after a multiple-testing correction

## Check your understanding (1/2)

- Which pricing restrictions follow from no arbitrage alone?
- Which of the restrictions used in the chapter depend on the Black–Scholes assumptions?
- Why can a delta-hedged option lose money even when the model is right?
- What role does the gap between realised and implied variance play in the hedging result?
- Does a fitted smile automatically exclude arbitrage?

## Check your understanding (2/2)

- How do you check the butterfly constraint for an SVI fit?
- Does a risk-neutral density forecast the real-world frequencies of returns?
- Why is the standard error $s/\sqrt{n}$ wrong for a mean over overlapping 21-day windows?
- Which standard error replaces it?
- How can two small $t$-statistics coexist with a large joint Wald statistic?

# Open Questions in Option Research (1/2)

- **Frontier**
  - Andersen, Fusari & Todorov (2015): a left-tail factor in index-option surfaces, not spanned by volatility, predicts future equity and variance premia
  - Bali et al. (2023): nonlinear machine learning on option and stock characteristics predicts equity-option returns out of sample
  - Open: jointly estimated, arbitrage-free surfaces with valid confidence bands; the pricing of 0DTE options
- **Next**: Chapter 13, machine learning for financial prediction
  - volatility and option-implied features among the inputs

# Open Questions in Option Research (2/2)

- **Further reading**
  - Textbooks: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), Gatheral (2006)
  - Pricing kernels: Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013); crypto options: Matic, Packham & Härdle (2023), Alexander & Imeraj (2020), Pele et al. (2023)
  - Quantinar: XFG Advanced Methods in Quantitative Finance, Pricing Kernels and Risk Premia, Hedging Cryptocurrency Options
  - Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_12
- Detailed examples and advanced topics are in Further Topics, after the Conclusions
  - the footer button leads there; the Back button returns to the starting slide

# The Practical Decision

- **What I choose**
  - for hedging: delta hedging at a frequency set by the error and the cost model; the s.d. of the error falls like $1/\sqrt{N}$ (weekly 21.5%, daily 9.5% of the premium)
  - for the smile: an SVI fit per expiry, accepted only after the butterfly and calendar checks; the density with bootstrap bands and the quoted range marked
  - for next month's variance: VIX$^2$, read with the joint Mincer–Zarnowitz test ($\hat{\beta} = 0.86$, unbiasedness rejected: $\chi^2_2 = 41.7$)
  - not the VRP as a return predictor after 2008: $R^2_{OS} = -22.2\%$, Clark–West $p = 0.26$
- **Under which conditions**
  - the density and the VRP are risk-neutral: prices of insurance, not forecasts of frequencies
  - overlapping 21-day windows: Newey–West with 21 lags and sensitivity to 42, 63 and 126 (standard error of the premium: 0.27 to 0.33 points)
  - mean premium: finite fourth moment of returns; coefficients and loss differences: finite second moment of the score or of the difference, and weak dependence (Chapter 1)
- **What result would make me drop it**
  - an SVI fit: $g(k) < 0$ or total variance falling with maturity; I then move to SSVI
  - the VIX as a forecast: past variance taking over its weight in the joint regression (here 0.08, SE 0.13, against 0.77)

# The Data: Volatility Indices and a Bitcoin Option Snapshot

| Series | What it measures | Period | Owner |
|---|---|---|---|
| S&P 500 | US equity index, daily close | 1990–2026 | S&P Dow Jones Indices |
| SPY | exchange-traded fund (ETF) on the S&P 500: open and close | 1993–2026 | State Street |
| VIX | 30-day implied volatility of the S&P 500 | 1990–2026 | Cboe |
| VIX9D, VIX3M | the same over 9 days and 3 months | VIX9D: 2011–2026; VIX3M: 2007–2026 | Cboe |
| VVIX | implied volatility of the VIX | 2006–2026 | Cboe |
| Bitcoin options | all listed contracts, one dated snapshot | 22 Sep 2026 | Deribit |
| DVOL | 30-day implied volatility of Bitcoin | 2021–2026 | Deribit |

- Daily data up to 18 September 2026
  - the option snapshot was taken on 22 September 2026, 07:51 UTC (UTC: Coordinated Universal Time)
- Implied volatility is quoted in annualised %; "vol points" = percentage points of volatility

# Notation: Prices, Expiry and Moneyness (1/2)

- $S_t$: price of the underlying at $t$; $K$: strike; $T$: expiry
  - $\tau = T - t$: time left, in years
  - $r$: continuously compounded risk-free rate
  - $q$: dividend yield
- $C$, $P$: prices of a call and a put
  - $(x)^+ = \max(x, 0)$
  - $\sigma_{imp}$: implied volatility (Section 6)
- European call and put pay off $(S_T - K)^+$ and $(K - S_T)^+$ at $T$
  - American options can be exercised at any time up to $T$
- Both pay-offs are convex in $S_T$, so an option's value increases with volatility (vega $> 0$, Section 4)

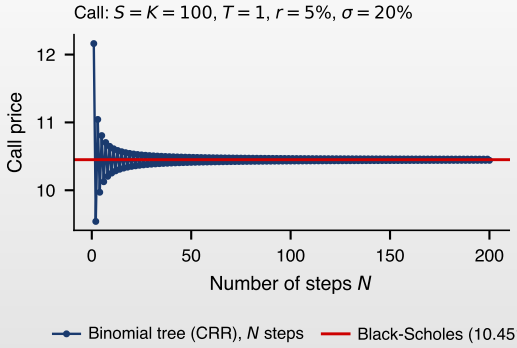
## Notation: Prices, Expiry and Moneyness (2/2)

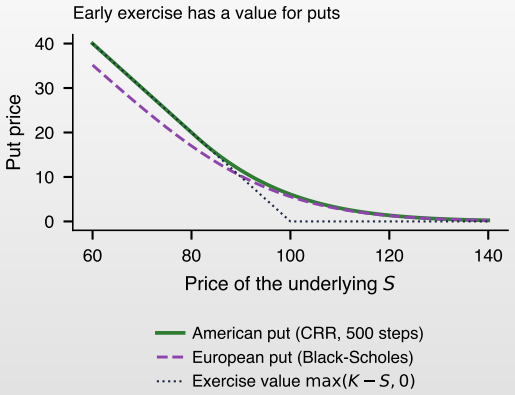
- Volatility $\sigma$: the annualised standard deviation of log returns $\ln(S_t/S_{t-1})$
  - Normal under Black–Scholes
- Used in most of the chapter: forward $F = Se^{(r-q)\tau}$
  - log-moneyness $k = \ln(K/F)$
  - total implied variance $w = \sigma_{imp}^2 \tau$
- ATM (at the money): $k \approx 0$
  - OTM (out of the money): puts with $K < F$ and calls with $K > F$
  - ITM (in the money): the opposite

## Context-dependent Symbols, Model-Free Bounds and Prerequisites

- $q(K)$: the risk-neutral density of $S_T$ (Sections 2 and 7)
  - in the binomial tree, $q$ is the risk-neutral up-probability (Section 3)
- $Q$: the pricing measure; $Q(K_i)$: the Cboe mid quote at strike $K_i$ (Section 8)
- Model-free bounds (no dividends): $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$ and $\max(Ke^{-r\tau} - S, 0) \leq P \leq Ke^{-r\tau}$
  - With $r > 0$ an American call on a stock without dividends is never exercised early: it is worth at least $S - Ke^{-r\tau} > S - K$
- Moneyness is measured here against the forward $F$
  - Seminar 12 and the smile chart of Section 6 use the spot $S$ (OTM call: $K > S$)
  - The two differ by the factor $F/S = e^{(r-q)\tau}$, close to 1 for short expiries: only strikes between $S$ and $F$ change class
- Assumed knowledge
  - Brownian motion, Itô's lemma and martingales (Chapter 11)
  - HAC (heteroskedasticity and autocorrelation consistent) standard errors (Chapter 8)

# From the Tree to Black–Scholes




MFM_ch12_binomial

## Convergence and the Early-Exercise Premium

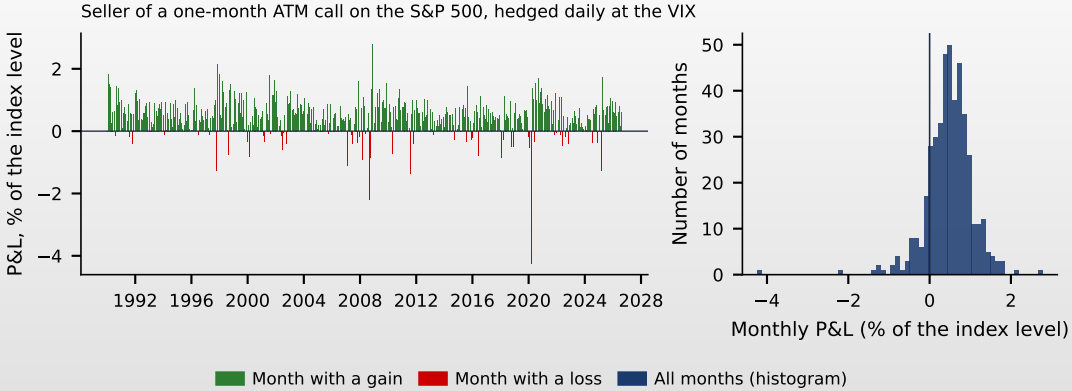
- Left: CRR price of a European call for $N = 1, \ldots, 200$ steps
  - $S_0 = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$
  - right: American vs European put
- Converges to Black–Scholes (10.451)
  - $N = 50$: 10.411; $N = 200$: 10.441; $N = 1000$: 10.449
  - The error is of order $1/N$ and oscillates: $N = 10$ gives 10.253, $N = 11$ gives 10.611 (strike on or between nodes; Seminar 12, A6)
- American put at $S = K = 100$: 6.090 vs European 5.574
  - early-exercise premium 0.516
  - deep in the money it pays to receive $K$ now

MFM_ch12_binomial

# Selling S&P 500 Volatility: the Strategy Tested

- The question: is implied volatility priced, on average, above realised volatility?
  - a sold, delta-hedged option answers it directly
- Every 21 trading days $t_0$: sell a synthetic 21-day ATM call ($K = S_{t_0}$)
  - priced by Black–Scholes with $\sigma = VIX_{t_0}/100$
  - the VIX comes from a strip of options, not the quoted implied volatility of the ATM call: the premium collected is set by construction
- Hedge it daily with $\Delta_i = \Phi(d_1)$ at the same volatility
  - interest and dividends are ignored ($r = q = 0$)
- Monthly P&L, in % of the index:
  - $\text{P\&L} = 100\left[C_0 + \sum_{i=0}^{20} \Delta_i \left(S_{t_{i+1}} - S_{t_i}\right) - \left(S_{t_{21}} - K\right)^+\right] / S_{t_0}$
- Why the VIX: it is the 30-day implied volatility available daily since 1990 in the course data
- Why ATM: vega is largest near the money, so the P&L mainly measures volatility

# Thirty-Six Years of Selling S&P 500 Volatility (1/2)




MFM_ch12_delta_hedged_sp500

# Thirty-Six Years of Selling S&P 500 Volatility (2/2)

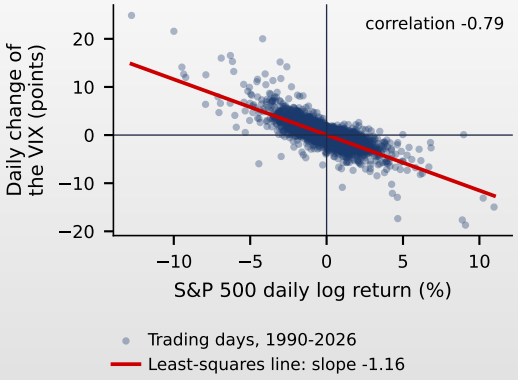
- Left: P&L of every month, in % of the index (green: gain, red: loss)
  - right: histogram of the monthly P&L
- 440 months from 1990-01: mean 0.48% a month, Newey–West $t = 16.0$ Newey & West (1987)
- Positive in 86% of months
- Worst month $-4.25\%$ (2020-03), skewness $-1.46$
  - many small gains, a few large losses
- Consistent with a **volatility risk premium**
  - On real option quotes, delta-hedged option gains are negative for the buyer Bakshi & Kapadia (2003)
  - here: a synthetic call priced at the VIX, without historical ATM quotes, bid–ask, dividends and financing; not the result of an executable strategy

MFM_ch12_delta_hedged_sp500

# Implied Volatility: Newton Inversion

- Newton's method: $\sigma_{k+\mathbf{1}} = \sigma_k - \dfrac{BS(\sigma_k) - C^{market}}{\mathcal{V}(\sigma_k)}$
  - $\sigma_k$: the $k$-th iterate; $\mathcal{V}(\sigma_k)$: vega at $\sigma_k$, the slope of the price in $\sigma$
  - Why Newton: vega is the exact derivative of the price in $\sigma$, so convergence is quadratic near the solution
- Example: $S = K = 100$, three months, $r = 2\%$, market price 4.50
  - start at $\sigma_{\mathbf{0}} = 20.00\%$
  - First step: $BS(\sigma_{\mathbf{0}}) = 4.2322$, $\mathcal{V} = 19.848 \Rightarrow \sigma_{\mathbf{1}} = 21.3495\%$
  - The second step changes $\sigma$ by less than $10^{-\mathbf{6}}$: $\sigma_{imp} = 21.3495\%$
- When vega is close to zero, a Newton step can leave the no-arbitrage range
  - vega is close to zero deep in or out of the money and at very short expiries
  - then use a bracketing solver

# The Leverage Effect in the Data




- Daily S&P 500 log return (%) vs the change of the VIX (index points) on the same day, 1990–2026
- Correlation $-0.79$ over 9 244 days
  - implied volatility rises when the market falls
- Slope $-1.16$ VIX points per 1% return
  - Steeper on down days ($-1.39$) than on up days ($-1.01$)
  - Mean VIX change: $+0.88$ on down days, $-0.76$ on up days
- The association is consistent with the leverage effect and motivates asymmetric models (Heston with $\rho < 0$, GJR-GARCH)
  - it does not estimate Heston's $\rho$ or the GJR coefficient directly: $\Delta VIX$ also contains changes in expectations and risk premia
  - GJR: Glosten–Jagannathan–Runkle, Chapter 5
  - Low-strike puts gain both from the fall and from the rise in volatility, hence they are expensive

MFM_ch12_leverage

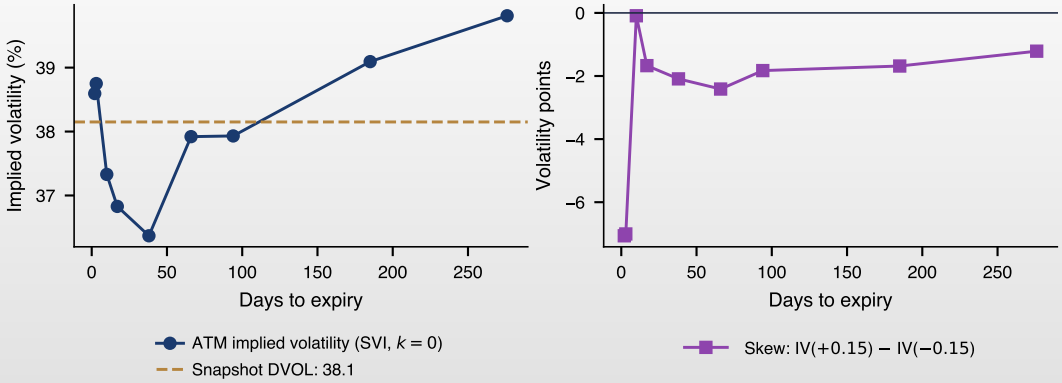
## Inverse Options: Why the USD Formulas Still Apply

- An inverse call pays $(S_T - K)^+ / S_T$ BTC at $T$
  - converted at $S_T$ USD per BTC this is exactly $(S_T - K)^+$ USD
  - the pay-off of a standard USD call
  - Example: $K = 90\,000$, $S_T = 100\,000$: the holder receives $10\,000/100\,000 = 0.1$ BTC, worth $10\,000$ USD
- Same pay-off, same value: $p \times S$ is the USD price of the standard call
  - a premium of $p$ BTC paid today costs $p \times S$ USD at today's index $S$
  - Black–Scholes/Black-76 and the exchange's mark IV (its reference implied volatility) therefore apply to $p \times S$ unchanged; this is the conversion used in the smile, density and VIX slides
- The rate: Deribit quotes a forward $F$ for each expiry
  - this chapter uses $e^{r\tau} = F/S$, the carry implied by the futures basis
  - in place of a Treasury rate
  - Check: an inverse put with $K = 90\,000$ ends at $S_T = 75\,000$. What does it pay in BTC and in USD? (Answer: 0.2 BTC, i.e. 15 000 USD)

## Are the Nine Bitcoin Fits Arbitrage-Free?

- For 2 October, 9 October and 27 November, $a$ hits its lower bound ($-1$)
  - the bound imposed in the optimisation
  - the individual parameters are not precisely identified
- Wings: largest slopes 0.51 (right) and 2.30 (left)
  - the left wing exceeds Lee's upper bound of 2
- Butterfly: $g(k) < 0$ somewhere on the grid $k \in [-1.5, 1.5]$ for 2 of 9 expiries
  - lowest $-0.017$, 2 October 2026
- Calendar: 1 of 8 adjacent pairs crosses
  - $w$(2 October 2026) $>$ $w$(9 October 2026) on 9% of $k \in [-0.5, 0.5]$
  - by at most 0.0018
  - Separate unconstrained fits can violate cross-expiry restrictions; these fitted curves do not establish arbitrage in executable market quotes
- Remedy: estimate the surface jointly under the constraints
  - SSVI, or least squares with $g \geq 0$ and $\partial_\tau w \geq 0$
  - also imposing the wing conditions

## Term Structure and Skew




MFM_ch12_btc_term

# Expiry, Skew and the DVOL Index

- Left: ATM implied volatility by expiry and the DVOL index
  - right: skew = IV($k = +0.15$)− IV($k = -0.15$)
  - IV: the implied volatility at that log-moneyness
- ATM between 36.4% and 39.8%; longest expiry (276 days) 39.8%
  - the curve falls at short maturities, then rises beyond about one month
- DVOL 38.1 vs ATM 36.8% and 36.4% around 30 days
  - a variance-strip index need not lie between ATM volatilities
- Skew closest to 0 at 10 days ($-0.1$), most negative at 66 days ($-2.4$), then $-1.2$ at 276 days
  - At 2 and 3 days, $k = \pm 0.15$ lies beyond the quotes (at most $k = 0.11$ and 0.14): the skew of about $-7$ points is an SVI extrapolation there
- Limitation: one snapshot is one day

MFM_ch12_btc_term

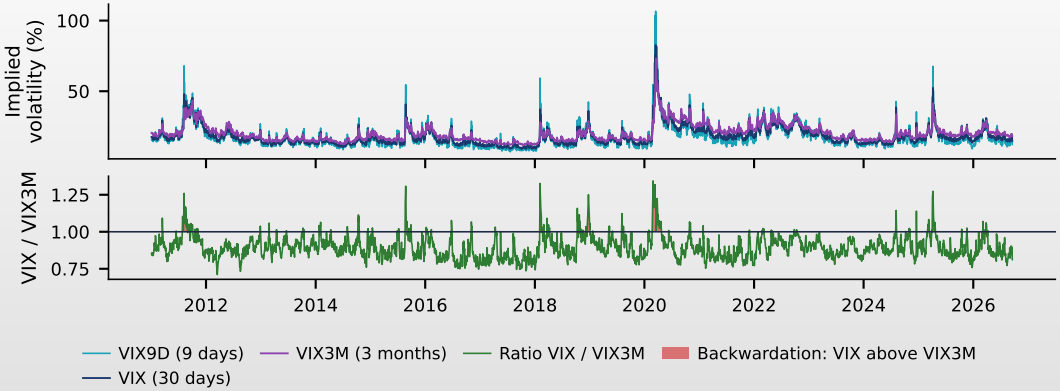
# The VIX Index: Computation Rules

- Cboe Volatility Index (VIX), published since 1993 Cboe (2025)
  - since 2003 computed from S&P 500 options with the Cboe formula
  - the series under this rule is computed back to January 1990, the start of the charts
- $\Delta K_i = (K_{i+1} - K_{i-1})/2$, one-sided at the two ends
  - at the lowest (highest) strike, $\Delta K$ is the full gap to the next higher (lower) strike
- Options with a zero bid are dropped, and each wing stops after two consecutive zero bids
- Two expiries around 30 days, interpolated to exactly 30 days; VIX $= 100\,\sigma$
- VIX futures since 2004, VIX options since 2006 Cboe (2025)
- The VIX is called the "fear gauge" Whaley (2000)

# Worked Example: a VIX for Bitcoin

- Cboe rules on our Deribit snapshot: USD mid $Q$ = BTC mid $\times$ index $S$
  - $F$ and $e^{RT} = F/S$ from Deribit's forward
  - Cboe: $F$ from parity, $R$ from Treasury yields
  - Near expiry: 17 days, 26 OTM options, $\sigma = 39.5\%$; next expiry: 38 days, 56 OTM options, $\sigma = 40.2\%$
- Interpolate the total variances $\sigma_i^2 T_i$ linearly in time to 30 days:
  - $\sigma_{30}^2 = \frac{1}{30}\left[T_1\sigma_1^2 \frac{T_2 - 30}{T_2 - T_1} + T_2\sigma_2^2 \frac{30 - T_1}{T_2 - T_1}\right]$ (days)
- Our Bitcoin VIX: 40.0; Deribit's DVOL at the same moment: 38.1 Alexander & Imeraj (2020)
  - both exceed the ATM volatilities
  - the variance swap also pays for the wings
- The gap to DVOL, 1.9 vol points, cannot be attributed to truncation
  - the SVI extrapolation it would require violates Lee's wing condition
  - Appendix; Seminar 12, A8

# The VIX Term Structure (1/2)



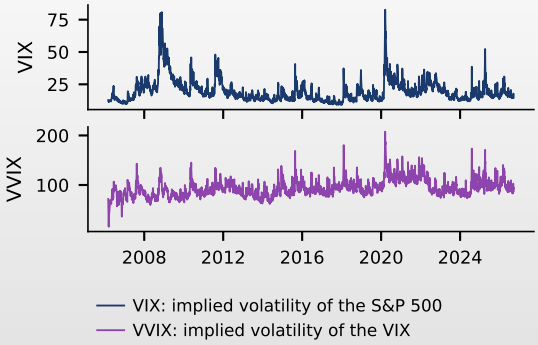

MFM_ch12_vix_term

# The VIX Term Structure (2/2)

- Top: implied volatility over 9 days, 30 days and 3 months
  - bottom: the ratio VIX / VIX3M, 2011–2026
- Normally upward sloping (**contango**): mean VIX / VIX3M = 0.89
  - VIX9D below the VIX on 73% of days
- Inverted (**backwardation**) on 7.6% of days, always in crises
  - fear is concentrated in the near term
- Over the next 21 days, after backwardation vs in contango:
  - S&P 500 log return 2.88% vs 0.80%; realised volatility 26.7% vs 13.6%
  - High risk and high expected return after panics Bollerslev, Tauchen & Zhou (2009)
- The comparison is descriptive
  - the 21-day windows overlap
  - backwardation days are few and clustered in a handful of crises

MFM_ch12_vix_term

# The Volatility of Volatility




- VIX (top) and VVIX (bottom), daily closes, 2006–2026
- Mean VVIX 93.5; mean daily ratio VVIX/VIX 5.3
  - Both are annualised implied volatilities, of percentage changes of the VIX and of the S&P 500: in relative terms the VIX is expected to move about 5.3 times more than the index
- VVIX peak 207.6 on 2020-03-16, the day of the VIX peak
- Daily changes of VIX and VVIX: correlation 0.71
- Volatility is itself random, the idea behind stochastic-volatility models such as Heston Heston (1993)

MFM_ch12_vvix

## Forecast-Evaluation Tools

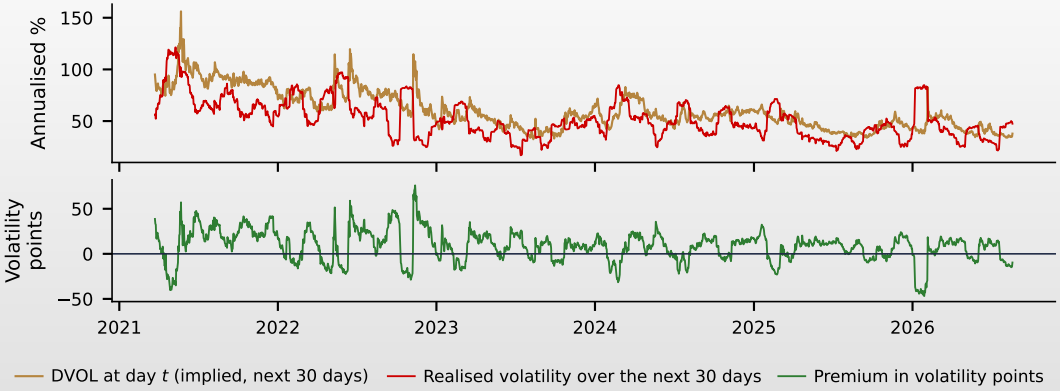
- **Instrumental variables**: the lagged $\text{VIX}^2$ instruments the $\text{VIX}^2$ measured with error
  - a large first-stage $t$ (regression on the instrument) means a strong instrument
- **HAR-RV**: forecast the realised variance of the next month
  - by a regression on its averages over the past day, week and month
- **QLIKE** loss $L = x - \ln x - 1$ with $x = RV/\hat{\sigma}^2$
  - zero for a perfect forecast
  - it punishes forecasts that are too low more than forecasts that are too high
  - Example: $RV = 100$; forecast 50 gives $x = 2$, $L = 0.31$; forecast 200 gives $x = 0.5$, $L = 0.19$
- **Diebold–Mariano**: a $t$-test that the mean loss difference between two forecasts is zero
  - with a HAC standard error
  - overlapping targets make the differences autocorrelated

# Logs, Instruments and HAR-RV: Inference Done Right

| Specification | $\hat{\beta}$ | SE | $t(\beta = 1)$ | Note |
|---|---|---|---|---|
| Levels, OLS | 0.860 | 0.105 | $-1.34$ | joint Wald $p = 8.9 \cdot 10^{-10}$ |
| Logs, OLS | 1.06 | 0.035 | 1.65 | intercept $-0.94$, $R^2 = 0.57$ |
| Logs, instrumental variables | 1.02 | 0.042 | 0.60 | first-stage $t = 37.7$ |

- Levels are dominated by 2008 and 2020
  - in logs the slope is close to 1
  - the premium is a proportional shift (intercept $< 0$)
- VIX$^2$ measures expected variance with error, which attenuates the ordinary least squares (OLS) slope
  - The instrument, VIX$^2$ 21 days earlier Christensen & Prabhala (1998), barely moves the slope
  - Valid only if the lag is uncorrelated with the forecast error (untestable with one instrument)
- Against HAR-RV Corsi (2009), in sample the HAR terms add information: $\chi^2_3 = 14.2$, $p = 0.003$
  - Out of sample from 2000: VIX about 6% better in QLIKE loss, Diebold–Mariano $p = 0.35$ Diebold & Mariano (1995): not significant
  - Test details: Appendix

# The Variance Risk Premium of Bitcoin (1/2)




MFM_ch12_vrp_btc

# The Variance Risk Premium of Bitcoin (2/2)

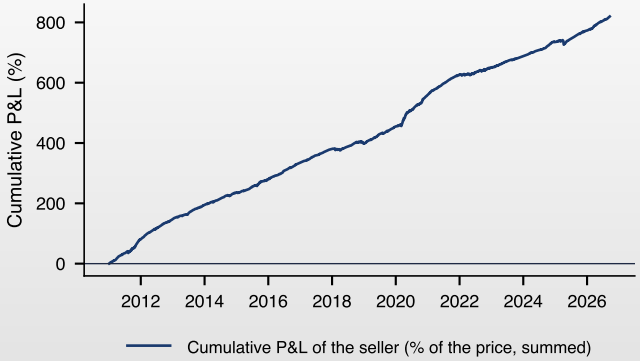
- Top: DVOL and Bitcoin volatility realised over the next 30 calendar days
  - Bitcoin trades every day
  - bottom: their difference, 2021–2026
- 1 975 days: mean DVOL 60.3 vs realised 51.7
  - Premium 8.5 vol points (SE 1.6, $t = 5.3$)
  - About 2.1 times the S&P 500 premium in vol points, but smaller relative to the level: 14% of mean DVOL vs 21% of mean VIX
- Positive on 71% of days, less often than the S&P 500
  - worst $-46.9$ vol points (window from 2026-01-28)
- Short sample: 66 non-overlapping 30-day periods Alexander & Imeraj (2020)

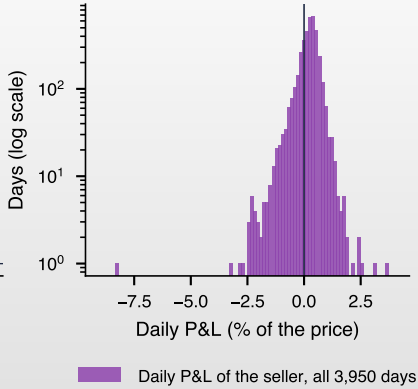
MFM_ch12_vrp_btc

# Selling a Same-Day Straddle: the Method

- The question: does the seller of an option expiring today also earn the volatility premium?
  - the premium of one-month options
- The course data contain no 0DTE prices: we build a stylised test
  - on SPY, 2011–2026 (VIX9D starts in 2011)
  - sell an ATM straddle (a call and a put with $K = S_t^{open}$) at the open
  - settle it at the close
- Premium: $\Pi_t = \sqrt{2/\pi}\,\sigma_{day}\,S_t^{open}$
  - the approximate Black–Scholes value of an ATM straddle
  - $E|Z| = \sqrt{2/\pi}$ for a standard normal $Z$
- $\sigma_{day} = \dfrac{VIX9D_{t-\mathbf{1}}}{100}\sqrt{\omega/252}$
  - the previous day's 9-day implied volatility, scaled to one session
- $\omega = 0.74$: the share of the session (open to close) in daily variance
  - estimated on 1993–2010, before the test period
- The seller's daily P&L, in % of the opening price:
  - $100\left(\Pi_t/S_t^{open} - \left|S_t^{close}/S_t^{open} - 1\right|\right)$

# Selling a Same-Day Straddle: a Stylised Test (1/2)




MFM_ch12_0dte_straddle

# Selling a Same-Day Straddle: a Stylised Test (2/2)

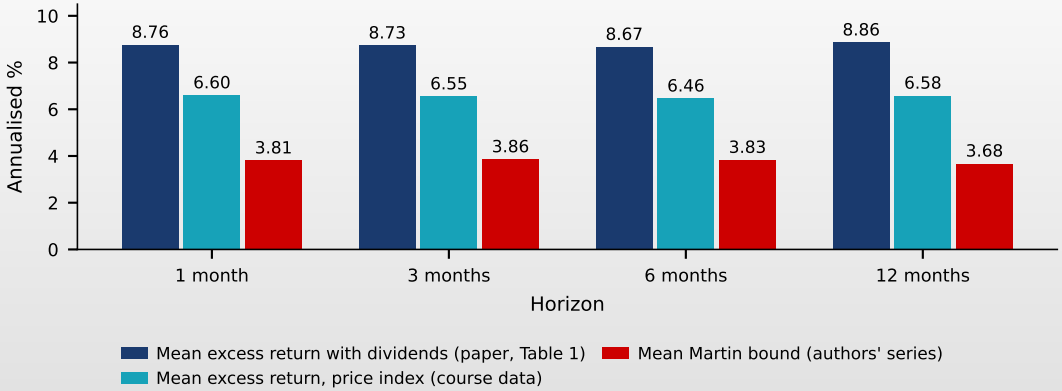
- Left: the seller's cumulative P&L (sum of daily P&L, in % of the price)
  - right: histogram of all daily P&L, log vertical axis
- 3 950 days: mean premium 0.76% vs mean absolute move 0.55%
  - the seller wins on 75% of days
  - 20.7 basis points a day
- Worst day −8.25% (2025-04-09), about 11 times the mean premium
  - stylised pricing measures the premium inside VIX9D, not traded 0DTE prices Bandi, Fusari & Renò (2023), Beckmeyer, Branger & Gayda (2023)

MFM_ch12_0dte_straddle

# Case Study: Results, Forecasting (Tables 5, 6 and 8)

- **Forecasting**: slopes positive but imprecise (Table 5)
  - With the nine predictors as controls: a one-s.d. rise in the 12-month bound, a 6.6-point higher 12-month excess return (Table 6)
  - Bootstrap $p$-values 20.7%, below 0.1%, 0.2%, 9.5%; with a Holm correction for the four horizons (Section 11), only 3 and 6 months stay significant at 5%
- **Out of sample**: the Martin bound loses to the historical mean at every horizon
  - the best case, Chabi-Yo–Loudis at 6 months, reaches 4.3%, not significant (Table 8)
  - low power is one possible explanation

# Case Study: Mean Premium and Mean Bound (1/2)




MFM_ch12_bck_bounds

# Case Study: Mean Premium and Mean Bound (2/2)

- Dark blue: Table 1 of the paper (S&P 500 with dividends)
  - red: mean of the authors' Martin bound
- Teal: course data (price index, the same formation days)
  - 7 789 formation days at 1 month, 7 598 at 12 months
- Mean slackness on the course data: 2.79, 2.69, 2.63, 2.91 points a year
  - at 1, 3, 6 and 12 months
  - vs 4.8–5.2 in the paper
  - The gap of 2.2–2.3 points is mostly the dividend yield missing from the price index
- The mean bound covers 41–44% of the premium with dividends and 56–59% of the price-index premium
- Single realisations fall below the bound on 41% of days at 1 month
- The average realisation never falls below the mean bound (Seminar 12, C3)

MFM_ch12_bck_bounds

## Case Study: an Open Extension

- The course data have VIX, not SVIX (Martin (2017), Result 3)
  - VIX weights OTM options by $1/K^2$
  - VIX exceeds SVIX when the left tail is priced
  - Is the Martin bound still valid on 2021–2026 (the 2022 bear market, the 0DTE era of Section 9)?
  - Is a VIX-based bound still valid?
  - If not, can the Kodde–Palm test detect it?

# Proof of Spanning, Moments and SVIX

- Proof for $S_T > F$, by parts:
  - $\int_F^{S_T} f''(K)(S_T - K)\,dK = f(S_T) - f(F) - f'(F)(S_T - F)$
  - the put integral vanishes
  - symmetric for $S_T < F$
- Take $e^{-r\tau}E^Q[\cdot]$ with $F$ the forward (a forward contract costs 0)
  - a bond, a forward and a strip of OTM options with weights $f''(K)$ replicate any smooth pay-off
- **Moments**: $f = [\ln(S/F)]^n$, $n = 2, 3, 4$, price the raw risk-neutral moments Bakshi, Kapadia & Madan (2003)
  - raw moments of the log return
  - centring and standardising gives volatility, skewness and kurtosis
- **SVIX** Martin (2017), from $f = S^2$:
  - $\mathrm{SVIX}_t^2 = \dfrac{2}{\tau R_f S_t^2}\left[\int_0^F P\,dK + \int_F^\infty C\,dK\right] = \mathrm{var}_t^Q(R_T/R_f)/\tau$
  - $R_T$, $R_f$: gross returns
  - $\mathrm{SVIX}^2$ gives the Martin lower bound on the equity premium (case study, Section 10)

Back

# Why $1/\sqrt{N}$? The Discrete Hedging Error

- Over one interval $\Delta t = T/N$: time decay against realised squared return
  - the hedged book earns $\frac{1}{2}\Gamma S^2\left[\sigma^2\Delta t - (\Delta S/S)^2\right]$
- Mean 0 under the model to leading order in $\Delta t$
  - $\mathrm{Var}[(\Delta S/S)^2] \approx 2\sigma^4\Delta t^2$
  - the terms are approximately uncorrelated
  - so $\mathrm{Var}(\text{error}) \approx \frac{1}{2}\sigma^4\Delta t\, E\int_0^T \Gamma_t^2 S_t^4\, dt$
- ATM call (derivation): $E[\Gamma_t^2 S_t^4] \approx \dfrac{K^2}{2\pi\sigma^2\sqrt{T^2 - t^2}}$
  - it integrates to $K^2/(4\sigma^2)$
  - so s.d. $\approx \sigma K\sqrt{T/(8N)} = \sqrt{\pi/4}\;\mathcal{V}\sigma/\sqrt{N}$
  - The last equality holds exactly for the vega at $d_1 = 0$, $\mathcal{V}_0 = K\sqrt{T}\,\varphi(0)$; here $d_1 = 0.15$ and $\mathcal{V} = 0.989\,\mathcal{V}_0 = 19.72$, so at $N = 12$ the two forms give 1.02 and 1.01
- Asymptotic theory of the replication error: Bertsimas, Kogan & Lo (2000)
  - it fails with jumps (the error does not vanish as $N \to \infty$)
  - with stochastic volatility (vega risk) and with costs

Back

# The Gamma Moment Behind the $1/\sqrt{N}$ Law

- ATM call, $S_0 = K$, $r = 0$, remaining time $\tau = T - t$
  - $\Gamma_t S_t^2 = S_t\,\varphi(d_1)/(\sigma\sqrt{\tau}) \approx K\varphi(d_1)/(\sigma\sqrt{\tau})$
  - with $d_1 \approx X/(\sigma\sqrt{\tau})$ and $X = \ln(S_t/K) \sim N(0, \sigma^2 t)$
- Gaussian integral: $E[\varphi(d_1)^2] = \dfrac{1}{2\pi} E\left[e^{-X^2/(\sigma^2\tau)}\right] = \dfrac{1}{2\pi}\sqrt{\dfrac{\tau}{\tau + 2t}}$, and $\tau + 2t = T + t$
- Hence $E[\Gamma_t^2 S_t^4] \approx \dfrac{K^2}{2\pi\sigma^2}\,\dfrac{1}{\sqrt{(T-t)(T+t)}} = \dfrac{K^2}{2\pi\sigma^2\sqrt{T^2 - t^2}}$, and $\displaystyle\int_0^T \frac{dt}{\sqrt{T^2 - t^2}} = \frac{\pi}{2}$ gives the integral $K^2/(4\sigma^2)$

Back

## The Moment Formula for the Wings

- Moment formula of Lee (2004): $\beta_R = \limsup_{k\to\infty} w(k)/k \in [0,2]$ and $\tilde{p} = \frac{1}{2\beta_R} + \frac{\beta_R}{8} - \frac{1}{2}$
  - $\tilde{p} = \sup\{p : E^Q[S_T^{1+p}] < \infty\}$: a steeper wing means fatter tails; the left wing mirrors it with negative moments
- Bitcoin: the largest right slope 0.511 gives $\tilde{p} = 0.54$
  - a formal extrapolation calculation only
  - this fitted slice fails a butterfly check, so it does not identify a valid risk-neutral moment
- For SVI a right-wing slope below 2 also gives the tail condition $d_+(k) \to -\infty$ of Gatheral & Jacquier (2014), Lemma 2.2

Back

## Estimators of the Risk-Neutral Density

- Fit a smile (SVI) and differentiate the implied call prices twice, as in the lecture
- Kernel regression of $C$ on $(K, \tau)$ Aït-Sahalia & Lo (1998)
  - the same under monotonicity and convexity in $K$ Aït-Sahalia & Duarte (2003)
- Smoothness-penalised implied trees Jackwerth & Rubinstein (1996); shape-invariant pricing kernels Grith, Härdle & Park (2013)
- Tails beyond the quotes: SVI wings or an extreme-value tail
  - either way an assumption, not an estimate

Back

## What the Strip Measures When Prices Jump

- With jumps $J$: the strip prices $\int \sigma_t^2 dt + 2\sum(e^J - 1 - J)$
  - but realised variance is $\int \sigma_t^2 dt + \sum J^2$
  - Gap $2E^Q \sum(e^J - 1 - J - J^2/2) \approx E^Q \sum J^3/3$: negative for crashes, so the strip understates expected realised variance Carr & Wu (2009)
- Merton model of Section 6 (one jump a year, mean $-8\%$)
  - strip 17.82 vs $\sqrt{E^Q[\mathrm{QV}]} = 18.00$ vol points, $-2.0\%$ in variance
  - QV: quadratic variation, the realised variance including jumps

Back

# Truncation and Discretisation of the Bitcoin Index

- Only a finite set of strikes is quoted Jiang & Tian (2005)
  - truncating the strip biases the index down
  - a coarse grid biases it up
- The Bitcoin 30-day index from the same two expiries, in three valid variants:
  - from quoted prices: 40.03
  - from SVI prices at the same strikes: 39.99
  - from SVI on a dense grid inside the quoted range: 39.77
- The gap of about 0.26 points between the three variants reflects smoothing and discretisation
  - all use the same quoted range: truncation is not quantified here
  - the omitted OTM contributions are positive, so they would not lower the index towards DVOL (1.9 points below)
  - quote selection and methodology remain
- Extending SVI beyond the quotes ($k \in [-4, 4]$) is not a fourth estimate
  - the left wing of the 17-day expiry has slope $2.30 > 2$ (Lee's bound)
  - so the extrapolated prices do not come from a valid density
  - the integral gives 128.47, a value with no economic meaning

Back

# Encompassing and Out-of-Sample Tests

- HAR-RV Corsi (2009): realised variance averaged over the last 1, 5 and 22 trading days
  - daily, weekly, monthly regressors
  - encompassing regression in logs Busch, Christensen & Nielsen (2011)
  - The target is $RV_{t,t+21}$ over the next 21 days: 22 is a look-back window, 21 the forecast horizon
  - VIX slope 0.90 (0.056); $R^2$: VIX 0.570, HAR 0.474, both 0.575
- Out of sample from 2000 (6 697 days)
  - coefficients re-estimated every 21 days on targets already observed
  - QLIKE ranks forecasts correctly with a noisy RV if RV is a conditionally unbiased proxy of the variance Patton (2011)
  - Loss ratio $\text{VIX}^2$ / HAR: 0.94, Diebold–Mariano $t = -0.93$ ($p = 0.35$) Diebold & Mariano (1995); VIX corrected by the Mincer–Zarnowitz regression: 0.95 ($p = 0.35$)
  - level forecasts from logs use $\exp(\hat{m} + \hat{s}^2/2)$, a log-normal correction, not a general identity
  - with smearing (mean of $e^{\hat{u}}$ on the training data): 0.95 ($p = 0.39$) and 0.94 ($p = 0.29$)
- Targets overlap by 20 of 21 days
  - the Diebold–Mariano variance is HAC with 21 lags
  - instrumental variables by just-identified generalized method of moments (GMM) with a HAC covariance

Back

# Does the Ex-Ante VRP Predict Returns? Overlapping-Horizon Inference

| Horizon | $\hat{b}$ | $t_{NW}$ | $t_{HH}$ | $t_{Hodrick}$ | $p_{boot}$ | $R^2$ (%) |
|---|---|---|---|---|---|---|
| $h = 1$ | 0.07 | 0.33 | 0.36 | 0.41 | 0.35 | 0.2 |
| $h = 3$ | 0.09 | 0.69 | 0.67 | 1.33 | 0.07 | 1.2 |
| $h = 6$ | 0.03 | 0.49 | 0.48 | 0.65 | 0.26 | 0.3 |
| $h = 12$ | −0.02 | −0.64 | −0.79 | −0.54 | 0.69 | 0.2 |

- Monthly 1990–2026: $\frac{12}{h}\sum_{j=1}^{h} r^e_{t+j} = a + b\,\mathrm{VRP}_t + u_t$ Bollerslev, Tauchen & Zhou (2009)
  - $\mathrm{VRP}_t = VIX_t^2/12 - RV_t$, ex-ante proxy with past $RV_t$ as the forecast
  - $r^e_{t+j}$: S&P 500 excess return of month $t + j$, $h$: horizon in months
- $VIX_t^2/12$ uses the month-end VIX
  - $RV_t$ is the sum of squared daily log returns (%) within month $t$

Back

# VRP: Inference with Overlapping Horizons

- If one-month errors are serially uncorrelated, overlap makes $u_t$ an MA($h-1$)
  - a moving average of order $h-1$
  - Newey–West ($t_{NW}$, $h$ lags), Hansen–Hodrick ($t_{HH}$) Hansen & Hodrick (1980), Hodrick 1B ($t_{Hodrick}$) Hodrick (1992): under the null its size is closer to the nominal level
  - In a given sample $t_{Hodrick}$ can still exceed $t_{NW}$: at $h = 3$, 1.33 vs 0.69
  - These bandwidths are choices, so check their sensitivity
- $p_{boot}$: one-sided, under $b = 0$ Stambaugh (1999)
  - VRP simulated as AR(1), a first-order autoregression ($\hat{\phi} = 0.13$)
  - residuals resampled jointly
- On the full 1990–2026 sample no horizon is significant
  - on 1990–2007 (Section 11) $h = 3$ is significant

Back

## Case Study, Filters and Forecasting Protocol

- Option data: Cboe for 1990–1995 and 2020–2021, OptionMetrics for 1996–2019
- Filters: standard options only (no weekly or quarterly), bid $> 0$
  - the OTM option at each strike
  - 7 to 550 days, at least 10 strikes per expiry
- Returns: S&P 500 with dividends minus the Fama–French risk-free rate
  - up to March 2021
- Out of sample (Table 8): monthly forecasts against the expanding post-1926 market mean
  - 60-month initial window
  - $R^2_{OS} = 1 - \sum \varepsilon_t^2 / \sum \nu_t^2$, $\varepsilon_t$ and $\nu_t$ the errors of the forecast and of the benchmark; Diebold–Mariano tests Diebold & Mariano (1995)

Back

# The Discovery Loop, Reusable Prompts (1/2)

- **Literature**:
  - `"List peer-reviewed tests of VRP return predictability since 2009: DOI, sample, horizon, table, headline estimate."`
  - Search with Semantic Scholar, Google Scholar or Elicit
- **Hypotheses**:
  - `"Give three competing explanations of VRP predictability and the testable implication of each for the horizon profile, for 2008-2026 and for non-US markets."`
- **Data and code**:
  - `"Write Python that builds VIX squared/12 and monthly realised variance from the daily VIX and S&P 500 closes of the course data and reproduces Table 2 of Bollerslev, Tauchen and Zhou (2009), h = 3, with Hodrick (1992) standard errors."`
  - Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot)

Back

# The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the results:
  - `"Draft a pre-analysis plan: horizons, out-of-sample start date, benchmark, test statistic and the multiple-testing correction across horizons."`
  - Placebo: VRP lagged 12 months or block-shuffled
  - Subsamples
  - Optional: 5-minute realised variance of SPY (from 2020), if you have intraday data, which the course data lack
- **Critique**:
  - `"Act as a hostile referee: why could this out-of-sample R2 be spurious or biased, and which test settles each objection?"`
  - Expect: the checks of the main slide, plus the bias with persistent regressors
- **Report**: every horizon and specification tried, in one table
  - with the plan dated before the results
  - AI use logged

Back

# Acronyms used in this material (1/2)

- **0DTE**: Zero Days To Expiry (options)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ATM**: At-The-Money: strike close to the forward, $k = \ln(K/F) \approx 0$
- **BTC**: bitcoin, currency code of the cryptocurrency
- **COVID-19**: COronaVIrus Disease 2019
- **CRR**: Cox–Ross–Rubinstein (binomial tree)
- **DVOL**: Deribit Volatility Index
- **ETF**: Exchange-Traded Fund
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GBM**: Geometric Brownian Motion
- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **GMM**: Generalized Method of Moments
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility
- **ITM**: In-The-Money: a put with $K > F$ or a call with $K < F$
- **IV**: Implied Volatility (mark IV; label of the Bitcoin term-structure chart), not instrumental variables
- **MA**: Moving Average

# Acronyms used in this material (2/2)

- **NBER**: National Bureau of Economic Research
- **OLS**: Ordinary Least Squares
- **OTM**: Out-of-The-Money: a put with $K < F$ or a call with $K > F$
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QV**: Quadratic Variation
- **RV**: Realised Variance
- **SE**: Standard Error
- **SSVI**: Surface SVI (Gatheral–Jacquier parameterisation of the whole surface)
- **SVI**: Stochastic Volatility Inspired (parameterisation)
- **SVIX**: Simple VIX: index of simple variance swaps (Martin, 2017)
- **US**: United States
- **USD**: US dollar, international currency code
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VAR**: Vector AutoRegression
- **VIX**: Cboe Volatility Index
- **VIX3M**: Cboe 3-Month Volatility Index
- **VIX9D**: Cboe 9-Day Volatility Index
- **VRP**: Variance Risk Premium
- **VVIX**: Cboe VIX of VIX Index
- **XFG**: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

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